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GnRH pulse generator activity in mouse models of polycystic ovary syndrome

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eLife Assessment

This **important** study reports findings on the GnRH pulse generator's role in androgen-exposed mouse models, providing further insights into PCOS pathophysiology and advancing the field of reproductive endocrinology. The experimental data were collected using cutting-edge methodologies and are **solid**. The findings, while interesting, are primarily applicable to mouse models, and their translation to human physiology requires cautious interpretation and further validation. This work will be of interest to endocrinologists and reproductive biologists.

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Abstract

One in ten women in their reproductive age suffer from polycystic ovary syndrome (PCOS) that, alongside subfertility and hyperandrogenism, typically presents with increased luteinizing hormone (LH) pulsatility. As such, it is suspected that the arcuate kisspeptin (ARN^{KISS}) neurons that represent the GnRH pulse generator are dysfunctional in PCOS. We used here *in vivo* GCaMP fiber photometry and other approaches to examine the behavior of the GnRH pulse generator in two mouse models of PCOS. We began with the peripubertal androgen (PPA) mouse model of PCOS but found that it had a reduction in the frequency of ARN^{KISS} neuron synchronization events (SEs) that drive LH pulses. Examining the prenatal androgen (PNA) model of PCOS, we observed highly variable patterns of pulse generator activity with no significant differences detected in ARN^{KISS} neuron SEs, pulsatile LH secretion, or serum testosterone, estradiol, and progesterone concentrations. However, an unsupervised machine learning approach identified that the ARN^{KISS} neurons of acyclic PNA mice continued to exhibit cyclical patterns of activity similar to that of normal mice. The frequency of ARN^{KISS} neuron SEs was significantly increased in algorithm-identified "diestrous stage" PNA mice compared to controls. In addition, ARN^{KISS} neurons exhibited reduced feedback suppression to progesterone in PNA mice and their pituitary gonadotrophs were also less sensitive to GnRH. These observations demonstrate the importance of understanding GnRH pulse generator activity in mouse models of PCOS. The existence of

cyclical GnRH pulse generator activity in the acyclic PNA mouse indicates the presence of a complex phenotype with deficits at multiple levels of the hypothalamo-pituitary-gonadal axis.

Introduction

Polycystic ovary syndrome (PCOS) is one of the most common causes of female infertility with variable reproductive and metabolic phenotypes (Bozdag, Mumusoglu et al. 2016, Lizneva, Suturina et al. 2016). Many women with PCOS have an elevated body mass index but up to 30% of patients do not, resulting in PCOS being further classified into obese and lean subtypes (Nestler and Jakubowicz 1997, Kumari, Haq et al. 2005, Toosy, Sodi et al. 2018). Although androgen excess, oligo-menorrhea/anovulation, and the existence of polycystic-like ovarian morphology are the key diagnostic indicators for PCOS, a common secondary feature is increased luteinizing hormone (LH) pulsatility (Waldstreicher, Santoro et al. 1988, The Rotterdam ESHRE/ASRM-sponsored PCOS consensus workshop group 2004). Basal and pulsatile LH secretion can be up to threefold higher in both lean and obese PCOS women and are considered to be a significant contributor to their subfertility (Morales, Laughlin et al. 1996, Dam, Roelfsema et al. 2002, Phylactou, Clarke et al. 2021). A presumed increase in the frequency of pulsatile gonadotropin-releasing hormone (GnRH) secretion in PCOS women results in an increased LH/follicle-stimulating hormone (FSH) ratio (Wildt, Hausler et al. 1981, Dalkin, Haisenleder et al. 1989). This is thought to drive excess ovarian androgen production that, in turn, impairs estrogen and progesterone feedback regulation of GnRH secretion (Pastor, Griffin-Korf et al. 1998, Chhabra, McCartney et al. 2005, Moore, Prescott et al. 2013, Moore, Prescott et al. 2015).

It has become clear that the arcuate nucleus kisspeptin (ARN^{KISS}) neurons represent the mammalian GnRH pulse generator (Clarkson, Han et al. 2017, Herbison 2018, Goodman, Herbison et al. 2022). These neurons exhibit brief episodes of synchronized activity that release kisspeptin on the distal projections of GnRH neurons to drive pulsatile GnRH secretion into the portal vasculature (Herbison 2021, Liu, Yeo et al. 2021). Recent studies indicate that the synchronized activity amongst ARN^{KISS} neurons required for pulsatile hormone secretion is generated by local glutamatergic transmission that is subsequently modulated by neurokinin B and dynorphin (Qiu, Nestor et al. 2016, Han, Morris et al. 2023, Morris and Herbison 2023). While the ARN^{KISS} neurons are the primary site of estrogen negative feedback suppressing pulsatile GnRH secretion (McQuillan, Clarkson et al. 2022), the mechanism of progesterone feedback remains unclear but could also involve the ARN^{KISS} neurons (Goodman, Holaskova et al. 2011). As such, it is thought that dysfunction within the ARN^{KISS} neuron pulse generator may be responsible for the elevated LH pulse frequency observed in PCOS (Walters, Gilchrist et al. 2018, Coutinho and Kauffman 2019, Rodriguez Paris and Bertoldo 2019, Ruddenklau and Campbell 2019).

We have developed GCaMP fiber photometry methodologies that allow the activity of the ARN^{KISS} neuron pulse generator to be monitored at high temporal resolution in freely behaving mice (Han, Kane et al. 2019, McQuillan, Han et al. 2019, Vas, Wall et al. 2024). Using this and other approaches, we report here a detailed investigation into the pulse generator activity in two widely used mouse models of PCOS including the peripubertal androgen (PPA) model of “obese PCOS”, and prenatal androgen (PNA) model of “lean PCOS” (Stener-Victorin, Padmanabhan et al. 2020).

Results

Peripubertal androgen (PPA) mice

1. Peripubertal dihydrotestosterone (DHT)-treated animals exhibit significantly increased body weight with acyclic estrous cycles

Female 129S6Sv/Ev C57BL/6 *Kiss1*$^{Cre/+}$,GCaMP6s mice exposed to DHT from 3 weeks of age (PPA: n=4) started to show a significant increase in weight 4 weeks after capsule implantation compared to controls with empty capsules (control: n=5) (Sidak's multiple comparisons test; Fig.1-Suppl. **Fig.1A**). The PPA animals were completely acyclic with vaginal cytology only ever showing the diestrous stage (Fig.1-Suppl. **Fig.1B-D**).

2. Peripubertal exposure to DHT results in a slowed GnRH pulse generator

Abrupt and large increases in arcuate nucleus kisspeptin (ARN^{KISS}) neuron population activity, termed synchronization events (SEs), are observed with GCaMP fiber photometry in mice (Clarkson, Han et al. 2017, Han, Kane et al. 2019, McQuillan, Han et al. 2019). Both adult PPA and control mice recorded in diestrus for 24 hours exhibited ARN^{KISS} neuron SEs (**Fig.1A,B**). The frequency of SEs in PPA mice (n=4) was approximately half that of controls (n=5) with SE frequency/hour showing an inhibitory trend (P=0.0714, Mann-Whitney U test; **Fig.1C**) and SE interval significantly increased (P=0.0159, Mann-Whitney U test; **Fig.1D**). The SE interval frequency distribution pattern revealed a shift to the right for PPA mice with a tight clustering within the 70-90-minute bin centers (**Fig.1E**). To compare the dynamics of SEs from control and PPA animals, continuous (10 Hz) photometry recordings were performed in diestrus to obtain high temporal resolution profiles (**Fig.1F**). These profiles were the same between control (n=5) and PPA (n=4) animals with SE duration, and both upswing and downswing durations, calculated from the full-width half maximum (FWHM) points, being not significantly different (**Fig.1G-H**).

Prenatal androgen (PNA) mice

1. The PNA mouse model exhibits highly disordered estrous cyclicity

Female mice originating from dams treated with DHT on gestational days 16, 17, and 18 were found to have a significantly increased anogenital distance (n=10; P<0.001, Mann-Whitney U test) compared with controls (n=9; **Fig.2A**). Body weight was not different between PNA and control animals at 10 weeks of age (n=12 per group; P=0.6602, Mann-Whitney U test; **Fig.2B**). The PNA mice exhibited very disordered estrous cycles remaining mostly in persistent diestrus (71 ± 3%) with occasional excursions to metestrus. The proestrus stage was never encountered and estrus was observed rarely (**Fig.2C-E**).

2. Circulating gonadal steroid hormone levels are not different between control and PNA animals

As androgen excess is one of the hallmarks of PCOS, we initially measured serum testosterone levels in control and PNA animals using a commercial testosterone enzyme-linked immunosorbent assay (ELISA) (Risal, Manti et al. 2021). This did not detect any difference in serum testosterone levels between control (n=13) and PNA (n=20) animals in diestrus (P=0.8203, Mann-Whitney U test; **Fig.2F**). Given concerns over the sensitivity and specificity of the steroid hormone ELISAs (Handelsman, Jimenez et al. 2015), we progressed to using ultrasensitive liquid chromatography-mass spectrometry (LC-MS) (Handelsman, Gibson et al. 2020) to measure the circulating levels of testosterone, androstenedione, estradiol, and progesterone in diestrous PNA and control mice (n=6 per group). No significant differences (Mann-Whitney U tests) were detected for androstenedione, estradiol, or progesterone while all samples were below the limit of detection for testosterone (< 0.01 ng/mL) (**Fig.2G-I**).

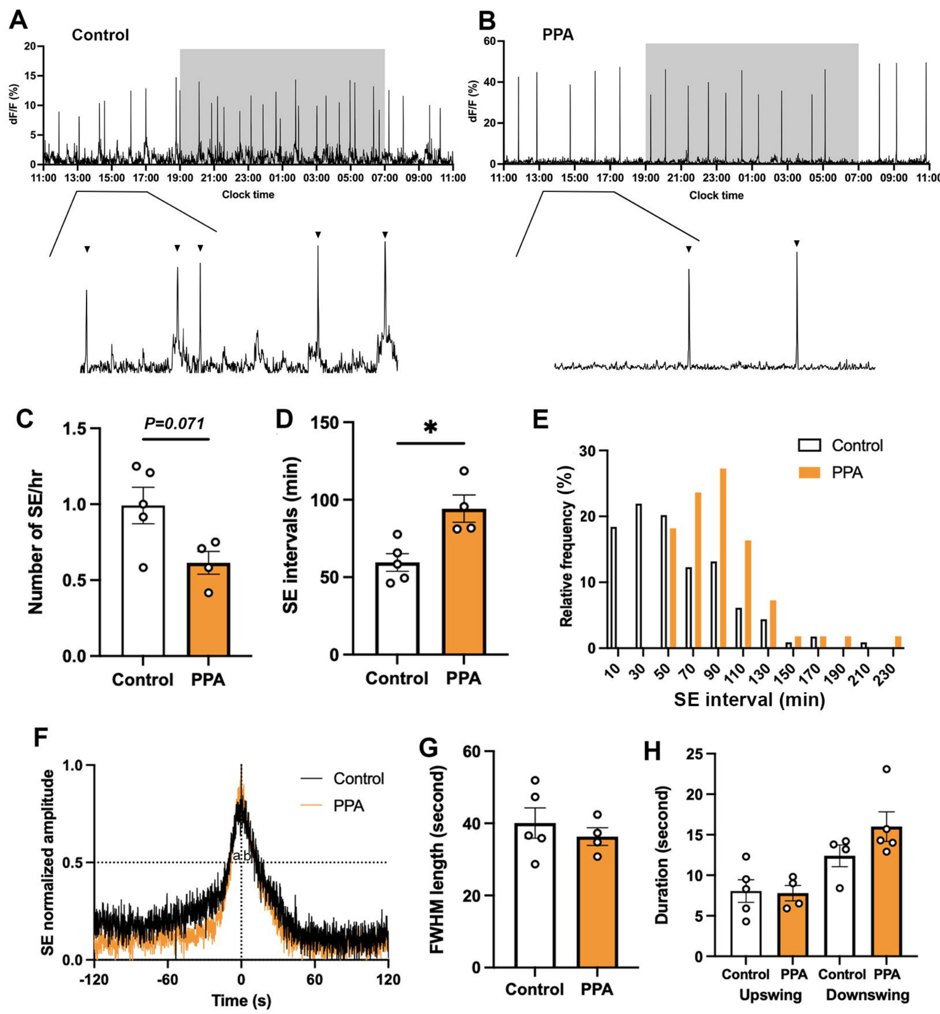


Figure 1.

Slower pulse generator activity in PPA animals

Representative 24-hour photometry recordings showing synchronization events (SEs) observed in diestrous (**A**) control and (**B**) PPA females with the light-off period (19:00-07:00) represented by the shaded area and expanded views of the traces (13:00-17:00) from (A-B). Triangles point to identified SEs. (**C**) SE frequency and (**D**) SE intervals in control (n=5) and PPA (n=4) mice. Mann-Whitney U tests. (**E**) Frequency histograms showing relative SE frequencies occurring in 20-minute bins, calculated separately for controls (white, n=114, 5 mice) and PPA (orange, n=55, 4 mice). X-axis represents the bin centers. (**F**) Continuous recordings at 10 Hz sampling rate showing normalized profile of SE overlaid from control (black, 7 SEs from 5 animals) and PPA (orange, 5 SEs from 4 animals). 'a' and 'b' give values of full width at half maximum (FWHM) upswing and downswing respectively. (**G**) FWHM length of control (n=5) and PPA (n=4) animals. Mann-Whitney U test. (**H**) Durations of upswing and downswing for control (n=5) and PPA (n=4) animals, respectively. Kruskal-Wallis test followed by Dunn's multiple comparisons test. Data show mean ± SEM. Each circle is an individual animal. * P<0.05

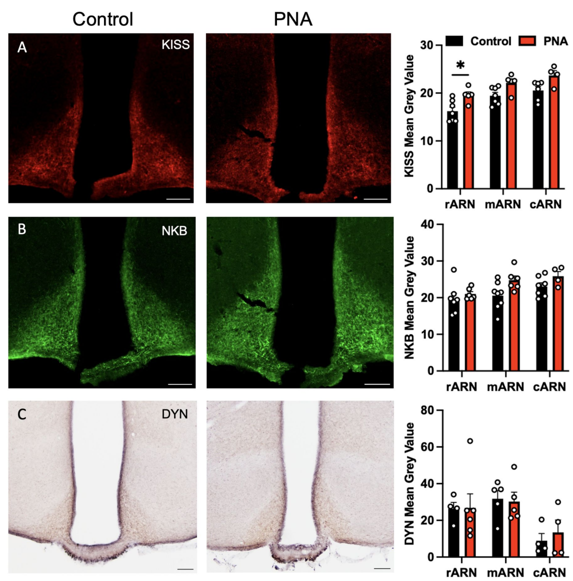


Figure 1 – figure supplement 1.

Peripubertal androgen (PPA) treatment causes an increase in body weight and a loss of estrous cyclicity

(A) Weekly body weight of animals (mean ± SEM) with blank (control: n=5) or dihydrotestosterone (PPA: n=4) capsules. Asterisks above each weekly data point denote significance from Sidak's multiple comparisons test. Asterisks between lines denote the mean effect of the capsule over 7 weeks. Two-way repeated-measure ANOVA with Sidak's multiple comparisons test. **(B)** The proportion of time spent at each stage of the estrous cycle over 21 days (n=4-5 per group). Two-way ANOVA followed by Sidak's multiple comparisons tests. **(C-D)** Representative graphs of estrous cycle pattern over 21 days in **(C)** control and **(D)** PPA females. P, proestrus; E, estrus; M, metestrus; D, diestrus. Data show mean ± SEM. Each circle is an individual animal. * P<0.05, ** P<0.01, *** P<0.001.

Figure 2.

Prenatal androgen exposure leads to increased anogenital distance and disrupted estrous cyclicity

(A) Anogenital distance (control: n=9; PNA: n=10) and (**B**) Body weight of control and PNA animals (n=12 per group). Mann-Whitney U tests. (**C**) Proportion of time spent at each estrous cycle stage over 21 days (n=7 per group). Two-way ANOVA followed by Sidak's post-hoc tests. (**D-E**) Representative graphs of estrous cycle pattern in (**D**) control and (**E**) PNA females. P, proestrus; E, estrus; M, metestrus; D, diestrus. (**F**) Serum testosterone level of control (n=13) and PNA (n=20) animals in diestrus measured by enzyme-linked immunosorbent assay (ELISA). The limit of detection for testosterone ELISA is 0.066 ng/mL. (**G-I**) Plasma levels of (**G**) androstenedione, (**H**) estradiol, and (**I**) progesterone in control and PNA animals measured using liquid chromatography-mass spectrometry (LC-MS). Blood samples were collected at 10:00 diestrus (n=6 per group). Dotted lines represent the limit of detection for each hormone. Androstenedione: 0.05 ng/mL; estradiol: 0.50 pg/mL; progesterone: 0.05 ng/mL. All samples were below limit of detection for testosterone LC-MS measurement (< 0.01 ng/mL). Mann-Whitney U tests. Data show mean ± SEM. Each circle is an individual animal. ** P<0.01, *** P<0.001.

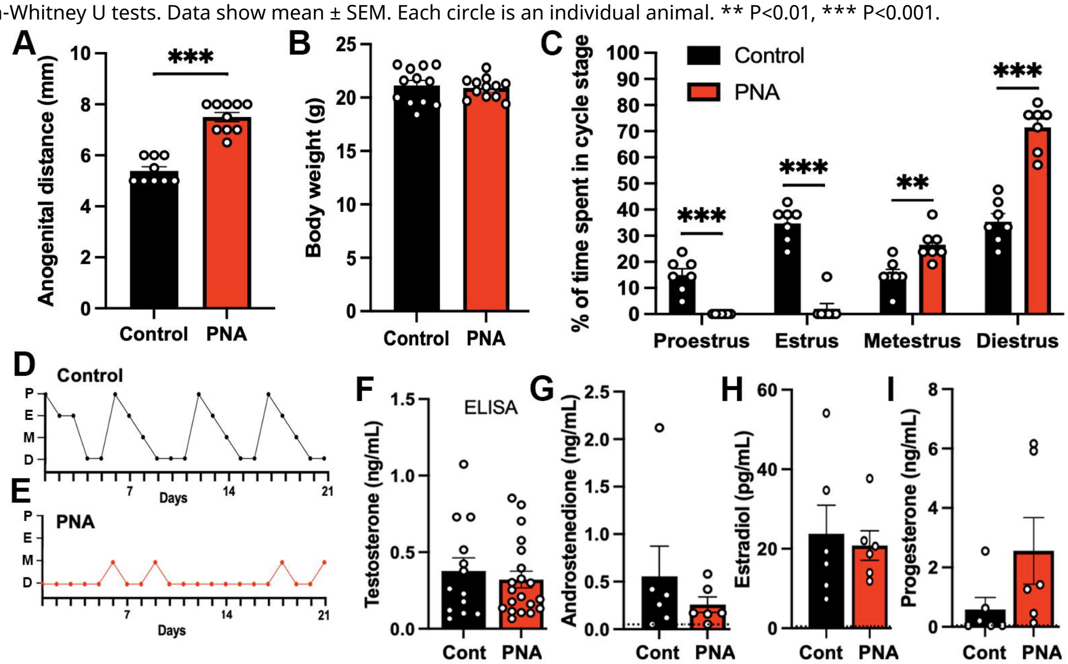
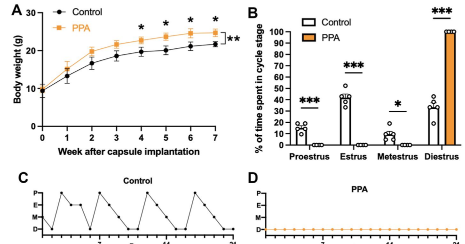


Figure 2 – figure supplement 1.

Slightly increased kisspeptin immunofluorescence in the rostral arcuate nucleus of PNA mice.

Representative arcuate nucleus images of (**A**) Kisspeptin (KISS), (**B**) Neurokinin B (NKB) immunofluorescent staining, and (**C**) Dynorphin (DYN) Nickel-3,3'-Diaminobenzidine (DAB) staining from control and PNA animals, along with the quantified immunoreactive level in rostral, middle, and caudal arcuate nucleus (rARN, mARN, cARN) in control (n=4-8) and PNA (n=4-6) animals. Scale bars represents 100 μm. Two-way ANOVA followed by Sidak's multiple comparisons test. Data show mean ± SEM. Each circle is an individual animal. Asterisk represents significant multiple comparisons results. * P<0.05



3. Kisspeptin, neurokinin B, and dynorphin immunoreactivity are relatively unchanged within the ARN of PNA mice

Previous studies have found variable effects of PNA treatment on kisspeptin, neurokinin B (NKB), and dynorphin protein and mRNA expression within the ARN (Barnes, Rosenfield et al., Yan, Yuan et al. 2014, Osuka, Iwase et al. 2016, Gibson, Jaime et al. 2021, McCarthy, Dischino et al. 2021, Moore, Lohr et al. 2021). To re-examine this issue, we assessed software-identified immunoreactive signal intensity for all three peptides in the ARN of control (n=4-8) and PNA (n=4-6) mice (Fig.2-Suppl. **Fig.1A-C**). This revealed a significant but modest elevation in kisspeptin immunofluorescence within the rostral ARN (P=0.0187, Sidak's multiple comparisons test) in PNA mice but not elsewhere in the ARN (Fig.2-Suppl. **Fig.1A**). No significant changes were detected for NKB or dynorphin immunoreactivity in any region of the ARN (Fig.2-Suppl. **Fig.1B,C**).

4. PNA animals exhibit highly variable patterns of ARN^{KISS} neuron SEs

Both control and PNA mice exhibited robust SEs across 24-hour recording periods (**Fig.3A-D**). Control mice recorded in diestrus had approximately 0.8 SEs per hour (**Fig.3E**) with an SE interval of 74.75 ± 5.71 min (**Fig.3F**) and a normal-like distribution of SE intervals (**Fig.3G**). Unexpectedly, diestrous PNA mice exhibited highly variable patterns of ARN^{KISS} neuron activity that could appear similar to diestrous controls (**Fig.3C**) or very different (**Fig.3D**) with episodes of multi-peak SEs (mpSEs). However, overall SE frequency and intervals were not different between PNA (n=6) and control (n=7) animals (**Fig.3E,F**; Mann-Whitney U tests). The frequency distribution of inter-SE intervals in PNA mice exhibited a peak interval cluster around the 10-minute bin center (39%) with a gradual tail off in SE intervals (**Fig.3G**). Continuous (10 Hz) photometry recordings in diestrus did not reveal any difference in the profiles of single SEs between controls (n=7) and PNA (n=6) animals (**Fig.3H-J**).

5. PNA mice have elevated mean LH secretion but normal LH pulse frequency

We have previously found a near perfect relationship between ARN^{KISS} neuron SEs and pulsatile LH secretion in male and female mice (Han, Kane et al. 2019, McQuillan, Han et al. 2019). To assess this relationship in PNA mice, ARN^{KISS} neuron recordings were commenced, and tail-tip blood samples taken at 12-minute intervals until an SE was detected after which the blood sampling rate was switched to 6-minute intervals. We observed a perfect correlation between SEs and pulsatile LH secretions in both control and PNA mice: every SE was followed by an LH pulse, and no LH pulses were detected without a preceding SE (**Fig.4A,C**). Temporal analysis of SEs and LH release indicated that the SE peak precedes each LH peak by 7.7 ± 0.6 minutes and 6.6 ± 0.4 minutes in control (n=7 from 4 mice) and PNA (n=7 from 5 mice) animals (P=0.190, Mann-Whitney U test), respectively (**Fig.4B,D**).

The absence of significant differences in ARN^{KISS} neuron SE frequency in PNA mice and their tight correlation with pulsatile LH secretion indicated that pulsatile LH secretion may not be different between PNA and control mice. Prior studies have found conflicting data with one finding no difference in LH pulse frequency (McCarthy, Dischino et al. 2021) and another reporting increased LH pulsatility (Moore, Prescott et al. 2015). To re-address this we undertook 4-hour tail-tip blood sampling at 10-minute intervals to assess pulsatile LH secretion in wild-type C57BL6/J diestrous-stage PNA (n=8) and control (n=7) mice. Mean LH levels were found to be significantly elevated in PNA mice (P=0.0205, Mann-Whitney U test; **Fig.4E-G**) but the variable LH pulse frequency (P=0.1843, Mann-Whitney U test) and amplitude (P=0.4630, Mann-Whitney U test) were unchanged compared to controls (**Fig.4H,I**).

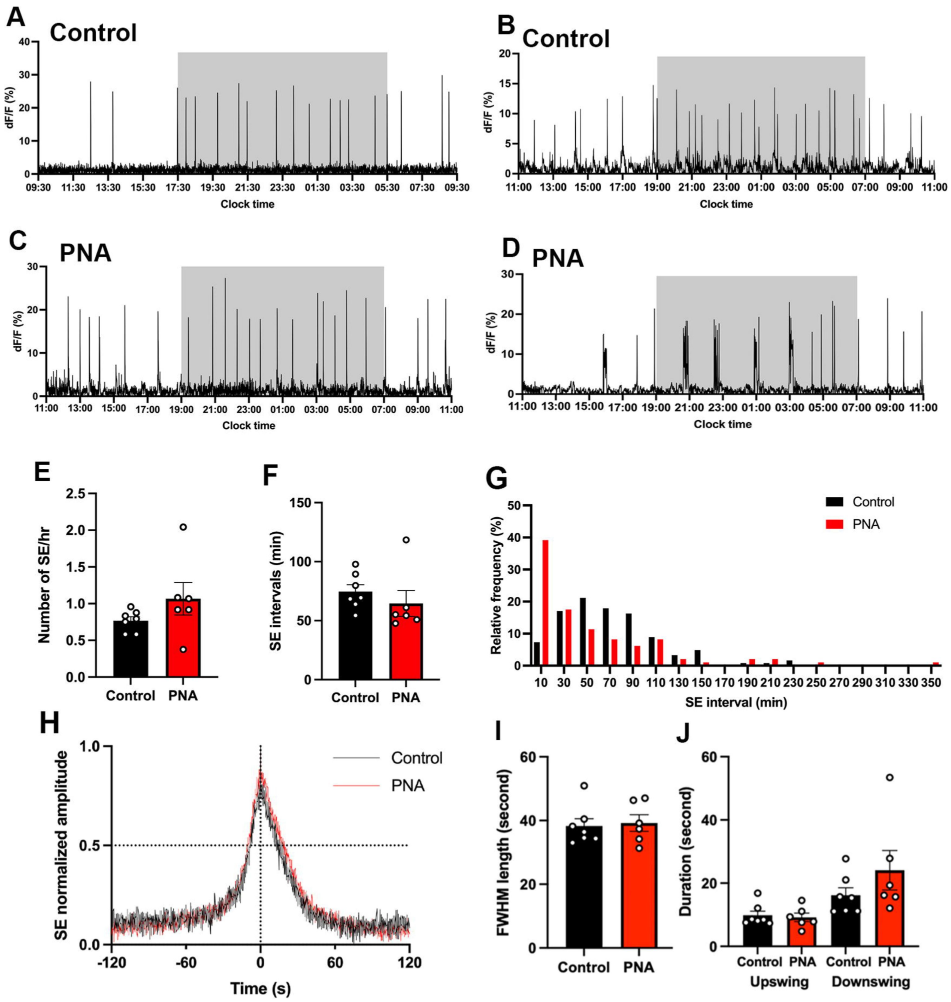


Figure 3.

PNA animals exhibit highly variable patterns of ARN^{KISS} neuron SEs

Representative 24-hour photometry recordings showing SEs observed in (**A,B**) two control and (**C,D**) two PNA females with the light-off period (17:30-05:30 or 19:00-07:00) represented by the shaded area. Recordings were performed when vaginal cytology indicated diestrus. (**E**) SE frequency and (**F**) SE intervals in control (n=7) and PNA (n=6) mice. Mann-Whitney U tests. (**G**) Frequency histograms showing relative SE frequencies occurring in 20-minute bins, calculated separately for controls (black, n=123, 7 mice) and PNA (red, n=98, 6 mice). X-axis represents the bin centers. Multi-peak SE (mpSE) with peaks occurring within 160 seconds was considered as one SE with interval calculated from the first peak. (**H**) Continuous (10 Hz) recording showing normalized mean profile of SE overlaid from control (black, 22 SEs from 7 animals) and PNA (red, 20 SEs from 6 animals). (**I**) FWHM length (upswing + downswing) in control (n=7) and PNA (n=6) animals. Mann-Whitney U test. (**J**) Durations of FWHM upswing and downswing for control (n=7) and PNA (n=6) animals, respectively. Kruskal-Wallis test followed by Dunn’s multiple comparisons test. Data show mean ± SEM. Each circle is an individual animal.

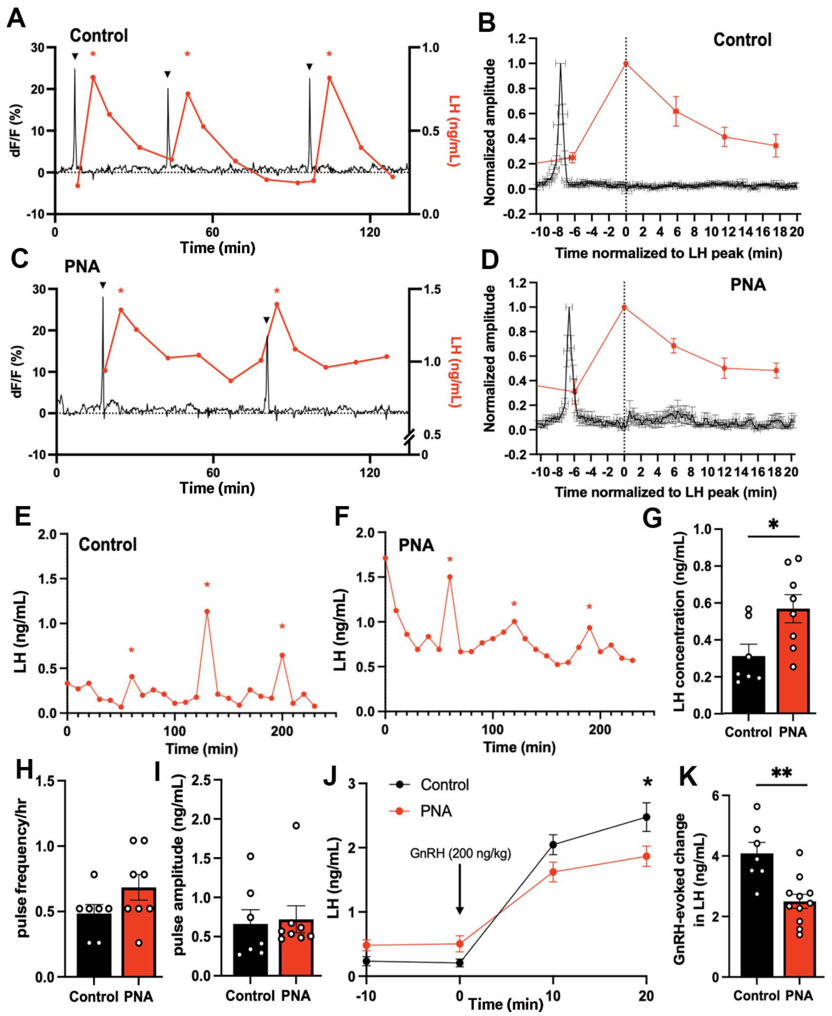


Figure 4.

Increased total LH concentration but normal LH pulse frequency in PNA animals

Representative examples from (**A**) control and (**C**) PNA mice showing the relationship of SEs (black) with LH secretions (red). Triangles and red asterisks indicate identified SEs and LH pulses, respectively. Normalized increase in LH plotted against the SEs in (**B**) control and (**D**) PNA animals. The amplitudes of SEs and serum LH levels were normalized to their peaks, with time 0 being the peak of LH in control (n=7 from 4 mice) and PNA (n=7 from 5 mice) animals. Representative examples of LH pulse profiles in (**E**) control and (**F**) PNA animals. Red asterisks indicate identified LH pulses. (**G**) Total LH concentration, (**H**) LH pulse frequency per hour, and (**I**) LH pulse amplitude in diestrous control (n=7, with 13 LH pulses) and PNA (n=8, with 21 LH pulses) mice during 4-hour sampling at 10-minute intervals. Mann-Whitney U tests. (**J**) Change in LH levels following an intraperitoneal injection of GnRH (200 ng/kg, i.p.) in diestrous control (n=7) and PNA (n=11) mice. An asterisk above the 20 minutes data point denote significance between LH levels in control and PNA from two-way repeated-measure ANOVA followed by Sidak's post-hoc tests. (**K**) GnRH-evoked changes in LH levels in control (n=7) and PNA (n=11) mice were calculated by the difference in LH levels before (−10 minutes and 0 minutes) and after (10 minutes and 20 minutes) GnRH injections. Mann-Whitney U test. Data show mean ± SEM. Each circle is an individual animal. * P<0.05, ** P<0.01

To evaluate whether the response of pituitary gonadotrophs to GnRH was altered, PNA and control mice were given GnRH (200 ng/kg, i.p.) at the end of the 4-hour tail-tip bleeding, and two further tail-tip blood samples collected for LH measurement at 10-minute intervals. Both control (n=7) and PNA (n=11) mice showed elevated LH levels in response to exogenous GnRH (**Fig.4J**). There was no significant main effect of PNA treatment (F(1,16)=0.6752, P=0.4233). However, PNA mice showed a small but significantly lower level of LH 20 minutes after GnRH injection (P=0.0152, Sidak's multiple comparisons test). The total GnRH-evoked increment in LH release was significantly lower in PNA mice compared to controls (P=0.0028, Mann-Whitney U test, **Fig.4K**).

6. The pulse generator exhibits cyclical activity in PNA mice

The patterns of ARN^{KISS} neuron SEs recorded from acyclic "diestrous" PNA mice were highly variable and could exhibit typical diestrous-like activity but also characteristics such as mpSEs (**Fig.3D**) that are only usually observed during estrus and metestrus (Vas, Wall et al. 2024). This suggested that, despite the acyclical state of the PNA mice, the pulse generator may still be undergoing cycles of activity. To assess this, we conducted long-duration photometry recordings in both PNA and control mice for four consecutive days starting on proestrus for controls and any day for the acyclic PNA mice. We have recently found that k-means clustering, an unsupervised machine learning-based approach, provides a very sensitive characterization of ARN^{KISS} neuron activity across the estrous cycle (Vas, Wall et al. 2024). This involves analysis of SEs based on five parameters: the frequency of regular single-peak SEs; the standard deviation of inter-SE intervals for regular SEs; and the number, duration, and profiles of mpSEs (**Fig.5A**).

All control animals showed typical cycling patterns of ARN^{KISS} neuron activity over the four-day recording period (n=5, two representative examples shown in **Fig.5C**). The most pronounced features of this cyclicity are the suppression of all SE activity during late proestrus to early estrus (represented by Cluster_0 activity) with a gradual return in SEs during the late estrus (Cluster_1). This is often followed by the appearance of mpSEs (Cluster_3/4) during the estrus to metestrus transition. The activity patterns then become the typical diestrous single SEs with variable frequencies (Cluster_1 & _2) (**Fig.5C**). Surprisingly, we observed that the ARN^{KISS} neurons in PNA mice also exhibited variable shifting SE profiles over the four-day recordings despite vaginal smears remaining in metestrus or diestrus throughout (n=4, two examples shown in **Fig.5D**). These PNA mice exhibited a cyclical pattern of activity reminiscent of control mice with a period of slow Cluster_0 activity directly followed by an mpSEs period (Cluster_3/4) and then regular diestrous activity (Cluster_1 & _2) (**Fig.5D**).

Overall, quantification of the time spent in each cluster pattern over the four days was remarkably similar between control (n=5) and PNA (n=4) mice (**Fig.5B**). The proportion of Cluster_0 (PNA 21%; control 21%) and Cluster_1 (PNA 44%; control 42%) activity was almost identical with the major trend being a nonsignificant increased time (P>0.05, Sidak's multiple comparisons test) spent in mpSE Cluster_3/4 activity (PNA 20%; control 12%) at the expense of the fast regular SE Cluster_2 pattern (PNA 15%; control 25%) (**Fig.5B**).

7. PNA mice have increased diestrous-like GnRH pulse generator frequency

Given that our earlier recordings from PNA mice (**Fig.3C,D**) were unavoidably taken from random times of the pulse generator cycle, we used the unsupervised machine learning algorithm above to identify "diestrous-like" (Cluster_1&_2) episodes of ARN^{KISS} neuron SE activity in PNA mice and compared this with diestrous controls (also defined by the same parameters). In this case, the frequency of SEs was 48% greater in PNA mice (n=6) compared to control animals (n=7; P=0.0029, Mann-Whitney U test; **Fig.6C**) as this was also reflected in a significantly lower inter-SE interval (P=0.0047, Mann-Whitney U test; **Fig.6D**). The frequency distribution of inter-SE intervals in PNA mice was slightly shifted to the left compared to control mice so that majority of PNA SE intervals were in 10-70 minute bin compared to the 30-90 minute bin centers in control mice (**Fig.6E**).

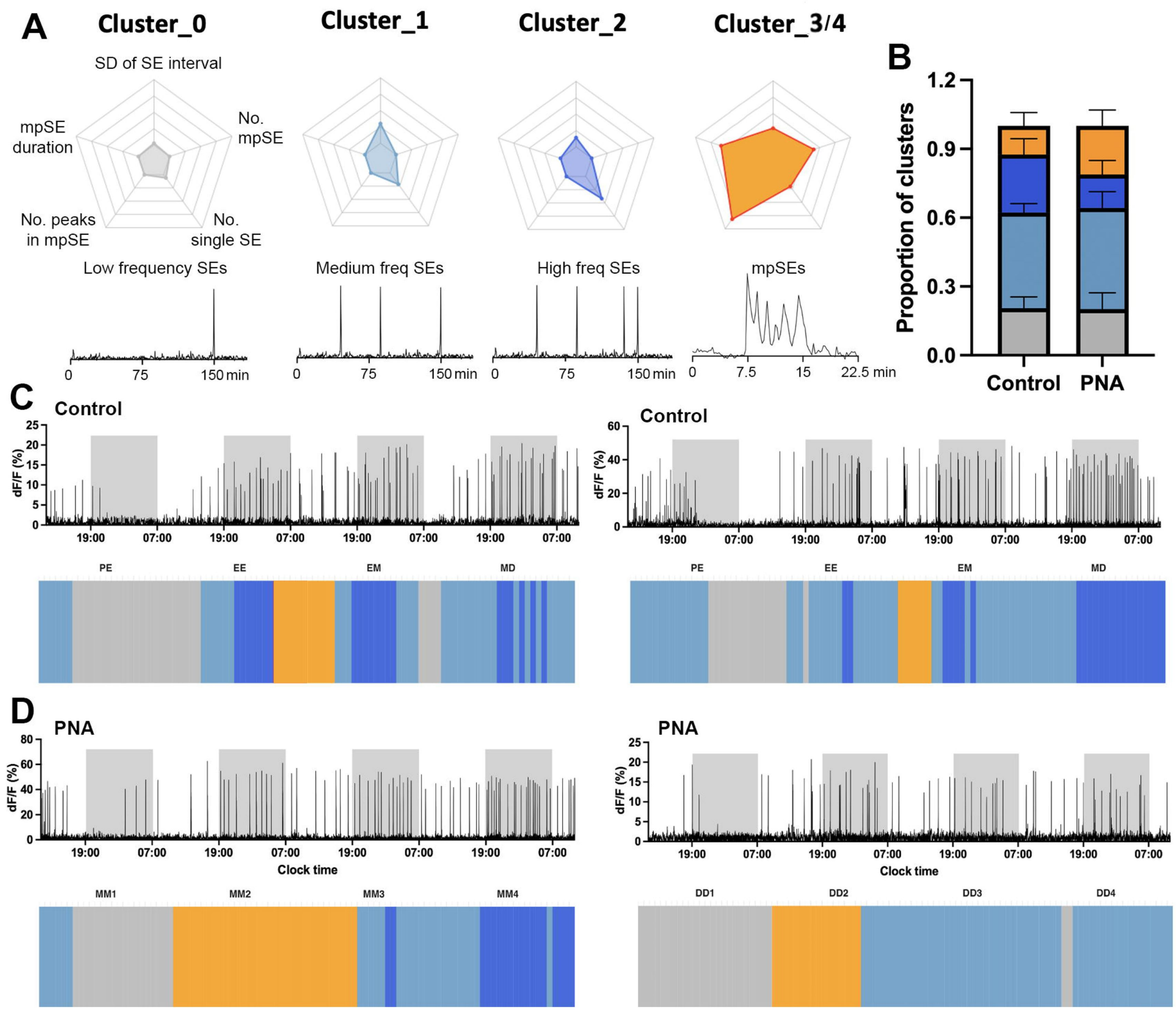


Figure 5.

PNA animals exhibit cycling pulse generator activity

(**A**). Cluster centroid values for normalized parameters used in k-means clustering with a schematic plot of corresponding photometry recordings patterns. The parameters used for cluster assignments are labelled in Cluster_0 with the following axes: (top) standard deviation (SD) of SE intervals, (top right) number of multi-peak SEs (mpSEs), (bottom right) number of single SEs, (bottom left) number of peaks in mpSEs, (top left) duration of mpSEs. All axes have a minimum value of 0 and maximum value of 1. (**B**) Bar graph indicating proportion of the cluster-assignments for 4-day recordings in control (n=5) and PNA (n=4) animals. Data show mean ± SEM. (**C-D**) Representative examples of four-day consecutive recording of ARN^{KISS} activity and corresponding hourly k-means cluster assignments in (**C**) two control animals starting in proestrus on day 1 and transiting to diestrus on day 4 according to vaginal smears; (**D**) two PNA animals remaining in either metestrus or diestrus for four days. Light-off periods (19:00-07:00) are represented by the shaded area in the photometry recording. Color fields represent the assigned clusters for the center (4^{th} hour) of the 7-hour moving time windows. Estrous stages of the mice determined by vaginal lavage at the beginning and the end of each 24-hour recording were labelled above the corresponding k-means cluster assignments: proestrus to estrus (PE), estrus to metestrus (EM), metestrus to diestrus (MD), stuck in metestrus (MM), or stuck in diestrus (DD). K-means clustering was performed without vaginal cytology information.

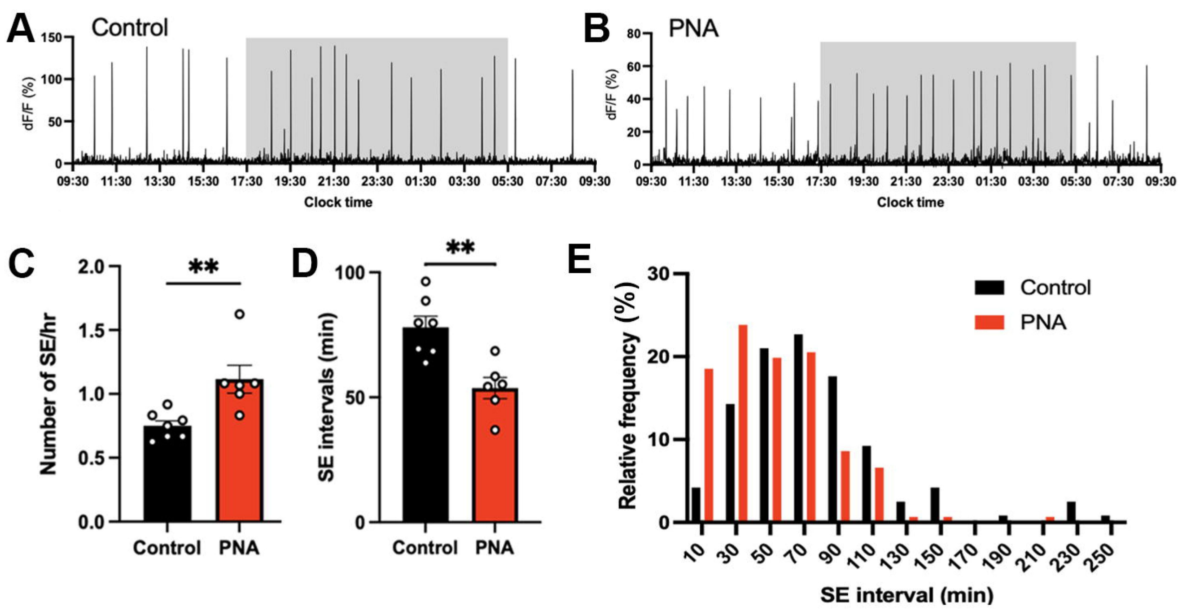


Figure 6.

Faster ARN^{KISS} neuron synchronization events in "diestrous-like" PNA animals

Representative 24-hour photometry recordings showing ARN^{KISS} neuron SEs observed in (**A**) diestrous control and (**B**) algorithm-identified diestrous PNA females. The light-off period (17:30-05:30) is represented by the shaded area. (**C**) SE frequency and (**D**) SE intervals in control (n=7) and PNA (n=6) mice. Mann-Whitney U tests. (**E**) Histogram showing percentage of SE interval frequencies occurring in 20-minute bins, calculated separately for controls (black, n=119, 7 mice) and PNA (red, n=151, 6 mice) animals. X-axis represents the bin centers.

8. PNA mice have impaired progesterone negative feedback

Progesterone negative feedback is thought to be abnormal in women with PCOS (Pastor, Griffin-Korf et al. 1998, Chhabra, McCartney et al. 2005) and PNA mice (Moore, Prescott et al. 2015). We assessed the response of ARN^{KISS} neuron activity to vehicle or progesterone (4 mg/kg, i.p.) in PNA and control mice with diestrous activity patterns. As expected (McQuillan, Han et al. 2019), we found a marked suppression of ARN^{KISS} neuron SEs (n=3, progesterone main effect $F(1,10)=7.333$, $P=0.022$) that lasted for up to 6 hours following progesterone in control mice (n=3, $P<0.001$, Sidak's multiple comparisons test, **Fig.7A,B,E**). The response in PNA mice was much less consistent, and overall, no significant differences in SE activity detected at any time point (n=3, progesterone main effect, $F(1,10)=1.917$, $P=0.196$, **Fig.7C,D,F**).

9. No differences exist in ARN^{KISS} neuron activity in ovariectomized (OVX) PNA and control females

Gonadal steroids play a key role in modulating the activity of the ARN^{KISS} neurons (McQuillan, Han et al. 2019, McQuillan, Clarkson et al. 2022). To assess whether any differences exist in ARN^{KISS} neuron activity in PNA mice in the absence of gonadal steroids, we undertook 24-hour fiber photometry recordings in mice that had been ovariectomized for at least 3 weeks. Removal of gonadal steroids results in increased SE frequency and the emergence of clusters of tightly coupled individual synchronizations as previously described (McQuillan, Clarkson et al. 2022) (**Fig.8A,B**). We observed no significant differences (Mann-Whitney U tests) between SE frequency (**Fig. 8C**) or intervals (**Fig. 8D,E**) in PNA and control animals (n=4 per group). Similarly, high-resolution recordings at 10 Hz did not reveal any differences in SE profiles between OVX control and OVX PNA mice (**Fig.8F,H**).

Discussion

It is important to establish animal models that faithfully recapitulate the symptoms and features of women with PCOS to be able to make progress in treating the disorder (Stener-Victorin, Padmanabhan et al. 2020). Increased LH pulse frequency is observed in ~75% of women with PCOS and is considered to be a significant contributor to their sub-fertility (Taylor, McCourt et al. 1997). We provide here a detailed assessment of the GnRH pulse generator in two commonly used mouse models of PCOS. The obese PPA mouse model was found to have a pulse generator that actually operates at a slower frequency than normal. In contrast, the PNA mouse model has a pulse generator that is operating faster than normal but, remarkably, maintains cyclical-like patterns of activity despite being in reproductively acyclic mice.

Pulse generator activity in PPA mice

The chronic administration of DHT to female mice from postnatal day 21 generates an androgen receptor-dependent mouse model of PCOS with infertility and metabolic disturbances (Caldwell, Middleton et al. 2014, Caldwell, Edwards et al. 2017, Stener-Victorin, Padmanabhan et al. 2020, Kerbus, Decourt et al. 2024). In agreement with those prior studies, we found that PPA mice start to exhibit increased body weight around four weeks after DHT exposure and are acyclic. Fiber photometry revealed that the frequency of ARN^{KISS} neuron SEs was reduced by ~50% in PPA mice compared to controls. This clearly indicates that the PPA mouse does not provide an appropriate PCOS model of elevated pulse generator activity. Given this, we did not proceed to examine pulsatile LH secretion in PPA mice but note that a recent study has reported pulsatile LH secretion to be unchanged in this animal model (Coyle, Prescott et al. 2022). It is very likely that chronic excess androgen activation in this model provides additional negative

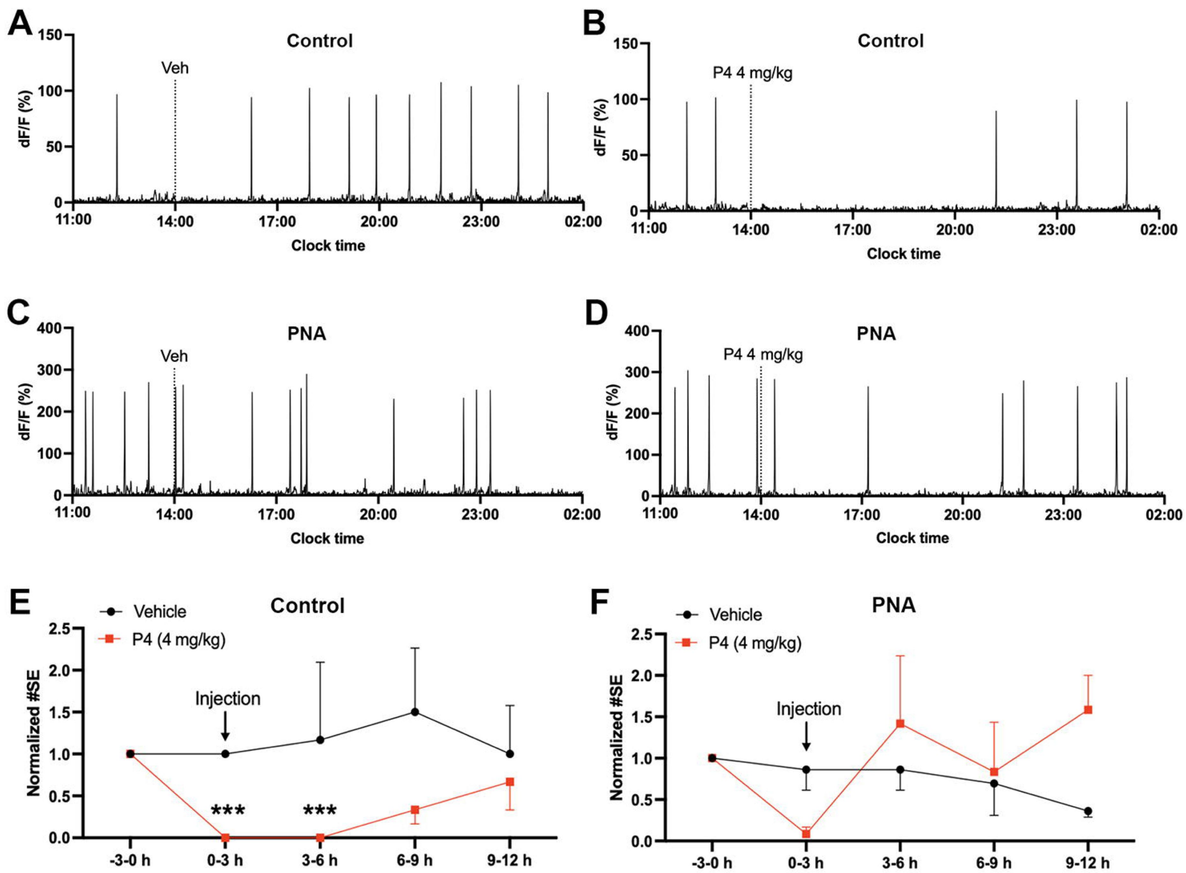


Figure 7.

Defective progesterone negative feedback in PNA animals

(**A-D**) Examples of 15-hour photometry recordings of the pulse generator activity in (**A-B**) control and (**C-D**) PNA animals receiving (**A**,**C**) vehicle (Veh) or (**B**,**D**) progesterone (P4, 4 mg/kg) i.p. injections. Dashed line represents when i.p. injection took place. Normalized number of SEs in (**E**) control (n=3) and (**F**) PNA (n=3) animals before and after vehicle (black) and P4 (red) injections. Data were analyzed in 3-hour bins and normalized to the number of SEs before injections (−3-0 h) for each mouse. The asterisks above the data point denote significant effects at indicated time periods compared to the pre-treatment period (−3-0 h) by Sidak's multiple comparisons test. Data represent mean ± SEM. Two-way repeated-measure ANOVA with Sidak's multiple comparisons tests. *** P<0.001.

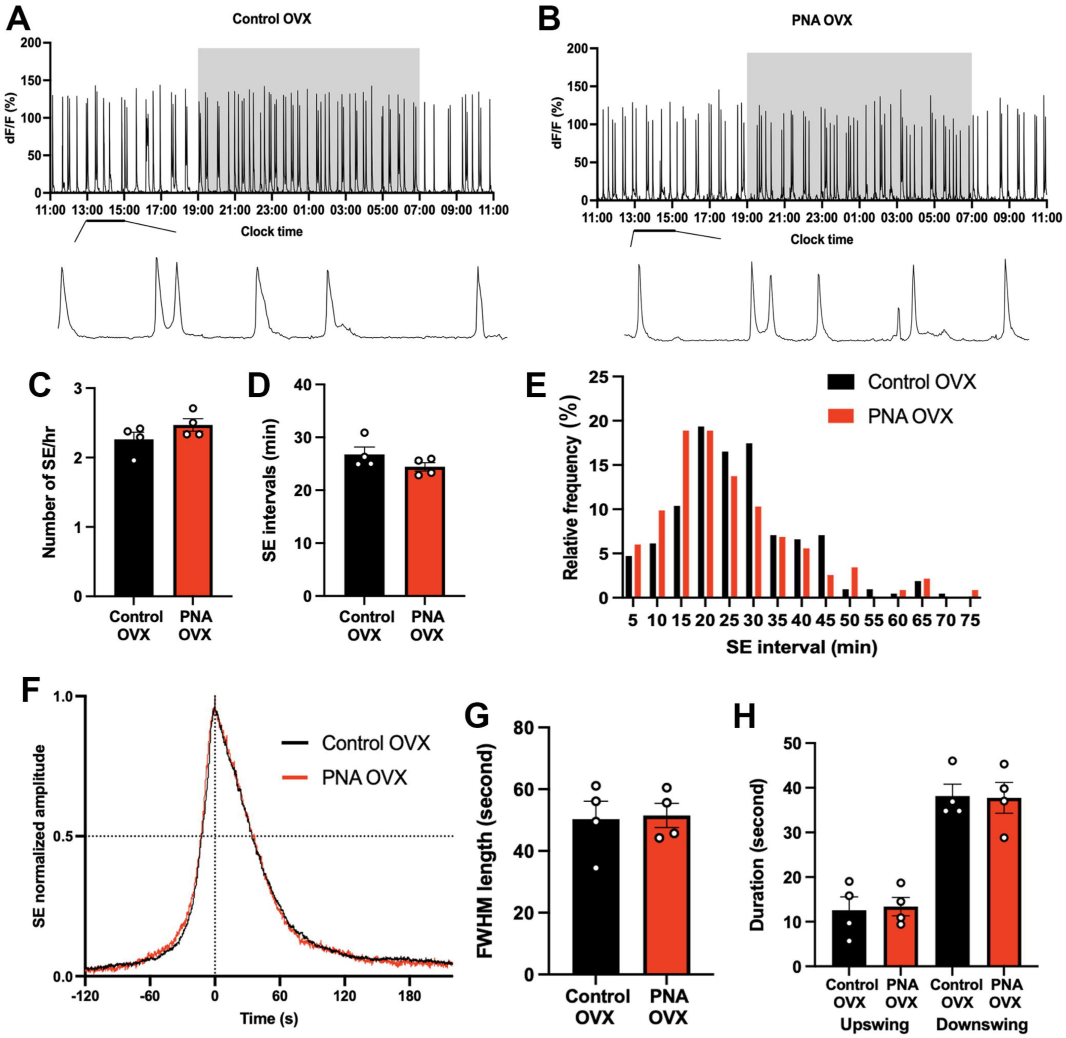


Figure 8.

Similar ARN^{KISS} activity and SE profiles in control and PNA animals following ovariectomy

Representative 24-hour photometry recordings showing ARN^{KISS} neuron SEs observed in (**A**) control and (**B**) PNA females 3 weeks after ovariectomy (OVX) with the light-off period (19:00-07:00) represented by the shaded area and expanded views of the traces (13:00-15:00) from (A-B). (**C**) SE frequency and (**D**) SE intervals in control and PNA mice (n=4), Mann-Whitney U tests. (**E**) Histogram showing relative SE frequencies occurring in 5-minute bins, calculated separately for controls (n=212, 4 mice) and PNA (n=233, 4 mice) animals. X-axis represents the bin centers. Each SE cluster was analyzed as a single SE with the interval calculated from the first peak. (**F**) Continuous recordings at 10 Hz sampling rate showing normalized profile of SEs following OVX overlaid from control (black, 14 SEs from 4 animals) and PNA (red, 10 SEs from 4 animals). (**G**) FWHM length of OVX control and PNA animals (n=4). Mann-Whitney U test. (**H**) Durations of upswing and downswing for control and PNA animals (n=4), respectively. P>0.05, Kruskal-Wallis test followed by Dunn's multiple comparisons test. Data show mean ± SEM. Each circle is an individual animal.

feedback suppression of ARN^{KISS} neurons. The treatment of adult OVX female mice with the same DHT regimen is reported to decrease LH pulse frequency and amplitude (Esparza, Terasaka et al. 2020).

Variable reproductive parameters of the PNA mouse model

The PNA mouse model has been studied by many different research groups with some features being identified in a highly consistent manner while others are not (Stener-Victorin, Padmanabhan et al. 2020). Common characteristics, also present in the current study, are those of increased anogenital distance, normal body weight, and highly disrupted estrous cycles (Sullivan and Moenter 2004, Silva, Prescott et al. 2018, Stener-Victorin, Padmanabhan et al. 2020, Gibson, Jaime et al. 2021, McCarthy, Dischino et al. 2021).

In contrast, reported differences in circulating gonadal steroid concentrations in PNA mice have been highly inconsistent among studies. Using ELISAs and radioimmunoassyays, some studies have demonstrated PNA mice to have elevated testosterone levels while others, including ourselves, find no differences (Sullivan and Moenter 2004, Roland, Nunemaker et al. 2010, Roland and Moenter 2011, Moore, Prescott et al. 2015, Lei, Ding et al. 2017, Silva, Prescott et al. 2018). It is unclear why this is the case, but we note that some studies have demonstrated a dependence on age. For example, Roland and co-workers reported testosterone levels to be normal in 5-month PNA mice but elevated at 8 months (Roland, Nunemaker et al. 2010) while Lei and colleagues found normal testosterone levels at ~1 month but elevated concentrations at 2-3 months (Lei, Ding et al. 2017). We assayed testosterone levels at 5-7 months of age in the present study.

Concerns over the specificity and selectivity of ELISAs for measuring gonadal steroids in mice (Handelsman and Wartofsky 2013, Auchus 2014, Wierman, Auchus et al. 2014, Handelsman, Jimenez et al. 2015) led us to re-assess gonadal steroid hormone levels using gold-standard ultrasensitive LC-MS (Handelsman, Gibson et al. 2020). We previously observed that testosterone levels were usually below the limit of detection in C57/BL6 female mice (Wall, Desai et al. 2023) but considered that any elevated concentrations in PNA mice might become detectable. Unfortunately, this was not the case, and overall it remains unclear if and when testosterone levels are elevated in PNA mice. Further detailed LC-MS assessment of testosterone levels will be required in the PNA model to address this important issue. We were, however, able to detect estradiol and progesterone concentrations and find that neither is different between control and PNA mice. A prior ELISA assessment also reported that estradiol and progesterone levels were unchanged in PNA animals (Moore, Prescott et al. 2015).

Another area of considerable inconsistency among studies involves the effects of PNA treatment on both mRNA and protein expression of kisspeptin, neurokinin B, and dynorphin in the ARN (Yan, Yuan et al. 2014, Osuka, Iwase et al. 2016, Gibson, Jaime et al. 2021, McCarthy, Dischino et al. 2021, Moore, Lohr et al. 2021). We re-investigated this question by quantifying the immunoreactive levels of these three proteins and observed an increase in only kisspeptin expression and only in rostral ARN of PNA animals. While there are important roles for these neuropeptides in modulating ARN^{KISS} neuron synchronizations (Han, Morris et al. 2023, Morris and Herbison 2023) and their subsequent activation of GnRH secretion (Liu, Yeo et al. 2021), these generally inconsistent findings indicate that any PNA-induced changes in the expression of the neuropeptides are subtle. Indeed, the principal determinant of kisspeptin, neurokinin B, and dynorphin expression in ARN^{KISS} neurons is estradiol (Navarro, Gottsch et al. 2009) and we demonstrate here that circulating estradiol levels are not altered in PNA mice.

Pulse generator activity and pulsatile LH in the PNA mouse model

One of the key features of the PNA mouse model is thought to be that of increased pulsatile LH secretion (Stener-Victorin, Padmanabhan et al. 2020). However, this has only been reported in a single study with PNA mice exhibiting an approximately 40% increase in LH pulse frequency (Moore, Prescott et al. 2015). A recent paper undertaking the same tail-tip pulse bleeding approach found that LH pulse frequency was not significantly altered in PNA mice (McCarthy, Dischino et al. 2021). Our present observations may explain these discrepancies. As the pulse generator continues to cycle in the acyclic PNA mouse, it is essentially impossible to define the "pulse generator stage" of a PNA mouse at the time of bleeding and consequently wide variations in LH pulse frequency would be expected. We find here that LH pulse frequency recorded from such random "pulse generator" cycling wild-type PNA mice is not significantly different to that of controls. Nevertheless, with the benefit of being able to measure ARN^{KISS} neuron activity directly, it has been possible to show that "diestrous-stage" PNA mice do have an increased rate of pulse generator frequency compared with controls. This may account for the overall increase in mean LH levels detected in the present study. As such, the PNA mouse model may to some degree replicate the increased pulse frequency seen in many women with PCOS.

The most striking feature of pulse generator activity in PNA mice is its ability to retain cyclical patterns of activity within a reproductively acyclic mouse. The pulse generator of PNA mice exhibits clear transitions from periods of "estrous-like" Cluster_0 activity through to mpSE-dominated Cluster_3/4 phase and onto regular "diestrous-like" Cluster_1&_2 patterns. Although the patterns of activity were not always identical, the overall percentage of time PNA mice spend in each pattern was not found to be significantly different to controls. We also note that the dynamics of individual ARN^{KISS} neuron SEs remain unchanged under all conditions in PNA mice, suggesting that the fundamental synchronization mechanism and numbers of contributing neurons is not altered (Han, Morris et al. 2023). This is consistent with the observation that PNA treatment does not alter ARN^{KISS} neuron firing activity in adult acute brain slices (Gibson, Jaime et al. 2021). Together, these observations indicate that the acyclic reproductive phenotype of PNA mice does not result from an absence of cyclical activity in the pulse generator. Although pulse generation in the mouse does not depend on GnRH neuron firing (Wang, Guo et al. 2020, Herbison 2021), it is noteworthy that a significant impact of PNA treatment on the activity of the GnRH neuron can be excluded (Jaime and Moenter 2022). The degree to which cyclical GnRH pulse generator activity in PNA mice reflects women with PCOS is unknown. However, a very wide range of LH pulse frequency is found in anovulatory women with PCOS (Taylor, McCourt et al. 1997), suggesting the intriguing possibility that the GnRH pulse generator activity may also continue to cycle in women with PCOS.

Gonadal steroid feedback in PNA mice

Pulse generator activity was found to be identical in ovariectomized PNA and control mice. This indicates that, despite androgen receptors being expressed by ARN^{KISS} neurons in the perinatal period (Watanabe, Fisher et al. 2023), excess DHT exposure at this time does not engender any fundamental differences in their synchronization behavior in the absence of gonads as adults. This suggests that the subtle increase in pulse generator frequency detected here in "diestrous-like" PNA mice arises from the effects of circulating gonadal steroids. As gonadal steroid levels are normal in PNA mice, it is likely that perinatal DHT exposure programs long-term changes in gonadal steroid receptor function in adult PNA mice. While direct estrogen feedback is critical in maintaining the suppressed activity of the pulse generator throughout the estrous cycle (McQuillan, Clarkson et al. 2022), ARN^{KISS} neuron ESR1 expression was not found to be altered in PNA mice (Moore, Lohr et al. 2021).

Reduced progesterone negative feedback is widely considered to be a major factor underlying enhanced LH pulsatility in women with PCOS (Pastor, Griffin-Korf et al. 1998, Chhabra, McCartney et al. 2005) and the ability of progesterone to suppress LH secretion is reduced in OVX PNA mice (Moore, Prescott et al. 2015). Whereas progesterone exerts a robust and prolonged suppression of ARN^{KISS} neuron SEs in normal (McQuillan, Han et al. 2019) and control mice, we find this to be much less effective in PNA females. Precisely how perinatal DHT treatment desensitizes ARN^{KISS} neurons to progesterone is unknown and could occur directly or indirectly (Moore, Lohr et al. 2021). Reported effects of PNA on progesterone receptors in the ARN are inconsistent with studies finding either reduced or unchanged expression (Moore, Prescott et al. 2015, Gibson, Jaime et al. 2021, Moore, Lohr et al. 2021). One recent study did find a 30% reduction in progesterone receptor mRNA expression by ARN^{KISS} neuron in PNA mice (Moore, Lohr et al. 2021).

Although we observed that the PNA mouse has reduced progesterone feedback at the level of the pulse generator, as is suspected for women with PCOS, it is unclear how this might pattern ARN^{KISS} neuron activity. Control and PNA mice exhibit similar proportions of time with halted or slow pulse generator activity (Cluster_0). This pattern of activity is thought to occur in response to increased progesterone level following the LH surge (McQuillan, Han et al. 2019, Vas, Wall et al. 2024). Thus, it is somewhat perplexing to find clear Cluster_0 activity in PNA mice that have reduced sensitivity to progesterone negative feedback. This might suggest that in other mechanisms can become responsible for luteal phase slowing of pulse generator activity in PNA mice. Equally, is unclear how reduced progesterone feedback could elevate the “diestrous-like” pattern of ARN^{KISS} neuron activity, although roles for progesterone outside the luteal phase have been suggested (Leipheimer, Bona-Gallo et al. 1984).

Previous studies have demonstrated that the estrogen-activated LH surge mechanism is normal in PNA mice (Moore, Prescott et al. 2013) and, despite proestrous smears never being observed in PNA mice, copra lutea can be found indicating that LH surges and ovulation can occur (Moore, Prescott et al. 2013, Lei, Ding et al. 2017, McCarthy, Dischino et al. 2021). This raises the intriguing possibility that the GnRH surge generator may also be operational in PNA mice but masked by abnormal downstream signalling. In that regard, we note here that pituitary gonadotroph sensitivity to GnRH is blunted in PNA mice as previously reported (Silva, Desroziers et al. 2019). Given that pulse generator activity is the same in ovariectomized PNA and control mice, reduced gonadotroph sensitivity would explain the reduced rise in LH following ovariectomy in PNA mice observed in other studies (Moore, Prescott et al. 2013, Moore, Prescott et al. 2015). Nevertheless, we note that the PNA mouse clearly does not model women with PCOS that show a much-elevated LH response to GnRH (Barnes, Rosenfield et al. 1989, Batrinos 1993, Morales, Laughlin et al. 1996, Patel, Coffler et al. 2003)

In summary, we report here that the GnRH pulse generator operates at a lower frequency in PPA mice, but at a higher frequency when “diestrous-like” pulse generator activity was analyzed in PNA mice. However, we note that many features of PNA mice such as their pulse generator free-running behavior in the ovariectomized state and gonadal steroid hormone concentrations are normal. Remarkably, we find evidence that the pulse generator continues to exhibit estrous cycle-like patterns of activity within an acyclic PNA mouse model. Thus, the anovulatory phenotype of PNA mice likely arises in a complex manner from deficits at many levels of the hypothalamo-pituitary-ovarian axis. The degree to which the PNA mouse model reflects women with PCOS remains an important, albeit very difficult, question to address.

Methods

Mice

Female 129S6Sv/Ev C57BL/6 *Kiss1*$^{Cre/+}$mice (Yeo, Kyle et al. 2016) were crossed with the Ai162 (TIT2L-GC6s-ICL-tTA2)-D Cre-dependent GCaMP6s males (JAX stock #031562) (Daigle, Madisen et al. 2018) to generate *Kiss1*-Cre/+,*Ai126D* mice as previously characterized (Han, Morris et al. 2023). Mice were group-housed in conventional cages with environmental enrichment under conditions of controlled temperature (22 ± 2 ^{0}C) and lighting (12-hour light/12-hour dark cycle; lights on at 05:30 or 07:00) with *ad libitum* access to food (RM3 (E), Special Diets Services, UK) and water. All animal experimental protocols were approved by the University of Cambridge, UK (P174441DE). Where required, mice were bilaterally ovariectomized (OVX) under isoflurane anesthesia at least 3 weeks prior to experimentation.

Peripubertal androgen (PPA) model

Peripubertal female mice were exposed to DHT as previously described (van Houten, Kramer et al. 2012, Caldwell, Edwards et al. 2017). Briefly, peripubertal female mice (21-23 days old) were implanted subcutaneously (s.c.) with either an empty (control) or dihydrotestosterone (DHT)-filled (PPA) 1-cm silastic implants (id, 1.47 mm; od, 1.95 mm, Dow Corning Corp, catalog no. 508–006). These silastic implants containing ~10 mg DHT provide steady-state DHT release for at least 6 months (Singh, O'Neill et al. 1995). The body weight of these animals was monitored weekly until optic fiber implantation at the 10 weeks of age.

Prenatal androgen (PNA) model

Adult females were paired with males and checked for copulatory plugs, indicating day 1 of gestation. Pregnant dams were given s.c. injections of sesame oil vehicle (100 µL) alone or containing 250 µg of DHT on gestational days 16, 17, and 18 (Sullivan and Moenter 2004). Female offspring from androgen-treated (PNA) and vehicle-treated (control) dams were used for experiments.

Estrous cycle determination

The estrous cycle stage was determined by the relative ratio of leukocytes, cornified, and nucleated cells observed in the vaginal epithelial cell smear samples on the morning of collection (Ajayi and Akhigbe 2020). The smears were collected using 5 µL of sterile phosphate-buffered saline (PBS) which was transferred to a glass slide to air dry before staining with filtered Giemsa (1:1 in MilliQ water) and examined under a light microscope.

Testosterone ELISA measurements and analysis

Blood samples for testosterone measurement were collected in the afternoon of diestrus from 22-30-week-old control and PNA animals through a tail-tip blood sample prior to ovariectomy or from the inferior vena cava before perfusion. The samples were then spun at room temperature at 3,000 rpm for 10 minutes. Serum was collected and kept at –20°C until the day of hormone measurement. Serum testosterone levels were measured using the testosterone rat/mouse ELISA kit according to the manufacturer's instructions (Demeditec Diagnostics, GmbH, DEV9911, Kiel, Bermany). The assay sensitivity for mouse testosterone ELISA was 0.066 ng/mL, and intra-assay coefficient of variation was 6.5%.

Liquid Chromatography–Mass Spectrometry (LC-MS)

Terminal blood samples were collected from the inferior vena cava of anaesthetized 18-22-week-old control and PNA animals at 10:00 on diestrus exactly as described previously (Wall, Desai et al. 2023). Plasma levels of estradiol, progesterone, testosterone, and androstenedione were measured using liquid chromatography-mass spectrometry (LC-MS) with detection limits of androstenedione: 0.05 ng/mL; estradiol: 0.50 pg/mL; progesterone: 0.05 ng/mL; testosterone: 0.01 ng/mL (Handelsman, Gibson et al. 2020).

Quantification of kisspeptin/neurokinin B/dynorphin expression in ARN

Adult diestrous PNA and control female mice were given a lethal dose of pentobarbital (3 mg/100μL, intraperitoneal, i.p.) and perfused transcardially with 4% paraformaldehyde (PFA). Brains were post-fixed in 4% PFA for 1 hour at 4°C and transferred to 30% sucrose at 4°C until sunk. Forty-μm-thick sections were collected with microtomes and processed for kisspeptin and neurokinin B (NKB) immunofluorescence and dynorphin (DYN) Nickel-3,3'-Diaminobenzidine (DAB) staining. For kisspeptin staining, brain sections were incubated with rabbit anti-kisspeptin (1:5,000, AC566; a gift from Dr. Alain Caraty) followed by goat anti-rabbit Alexa Fluor 568 (1:500, Molecular Probes). For NKB staining, brain sections were incubated with guinea pig anti-NKB (1:5,000, IS-3/61, a gift from Dr. Philippe Ciofi) followed by biotinylated goat anti-guinea pig IgG (1:1,000, Vector Laboratories) and DyLight Streptavidin 488 (1:500, BioLegend). Images were taken using a Leica SP8 Laser Scanning Confocal Microscope (Leica Microsystems) at the Cambridge Advanced Imaging Center and analyzed using ImageJ. Brain sections for DYN Nickel DAB labeling were incubated with rabbit anti-DYN B (1:12,000, a gift from Dr. Philippe Ciofi) (Griffond, Deray et al. 1993) followed by biotinylated goat anti-rabbit IgG (1:200, Vector Laboratories). Brain sections were further incubated in avidin-biotin peroxidase solution before reacting with Nickel DAB developing solution. Images were taken using an Olympus BX43 upright microscope with DP74 digital camera (Olympus Life Science) and analyzed using Image J.

Stereotaxic surgery and GCaMP fiber photometry

Adult female PNA, PPA, and their respective control *Kiss1*-Cre/+,*Ai126D* mice (10-14 weeks old) were anesthetized with 2% isoflurane, given meloxicam (5 mg/kg, s.c.), buprenorphine (0.05 mg/kg, s.c.), and dexamethasone (10 mg/kg, s.c.) and placed in a stereotaxic apparatus. A small hole was drilled in the skull and a unilateral indwelling optical fiber (400 μm diameter; 0.48 NA, Doric LenSEs, Quebec, Canada) was implanted directly above the mid-caudal ARN (2.0 mm posterior to bregma, 3.0 mm lateral to the superior sagittal sinus, 5.93 mm deep). Following optic fiber implantation, mice were monitored through recovery for 5 days and received post-operative analgesics (meloxicam, 5 mg/kg; orally) for up to two days. Then mice were handled daily for at least 3 weeks and habituated to the fiber photometry recording setup before fluorescence signals were recorded from freely behaving mice in their home cages.

Fiber photometry was undertaken as detailed previously (Clarkson, Han et al. 2017). Photometry systems were built using Doric components (Doric Lenses, QC, Canada) and National Instrument data acquisition board (TX, USA) based on a previous design (Lerner, Shilyansky et al. 2015). Blue (465-490 nm) and violet (405 nm) LED lights were sinusoidally modulated at frequencies of 531 and 211 Hz respectively. Both lights were focused onto a single fiber optic connected to the mouse. The light intensity at the tip of the fiber was 30-80 micro watts. Emitted fluorescence signal from the brain was collected via the same fiber, passed through a 500-550 nm emission filter, before focused onto a fluorescence detector (Doric, QC, Canada). The emissions were collected at 10 Hz and the two GCaMP6 emissions were recovered by demodulating the 465-490 nm signals (calcium-dependent) and 405 nm (calcium independent) signals.

Fluorescence signals were sampled using either a scheduled mode (5/15 seconds on/off) or a continuous mode of light emission with custom software (Tussock Innovation). Twenty-four-hour recordings were made from PNA, PPA, and their respective control female mice in the diestrous stage starting 4 hours after lights-on. Four-day recordings were started on proestrus for control mice and on any diestrous or metestrous stage for PNA mice. Continuous data acquisition mode was used to provide high temporal resolution comparisons of SEs. Signals from each SE were normalized to its peak (intact: −120 to +120 seconds; OVX: −120 to +200 seconds from the peak time). The full widths at half maximum (FHWM) of SE were determined at half-maximal amplitude of the normalized SEs. Averaged SE parameters for each animal were calculated before computing the means for each treatment group.

Analysis was performed in MATLAB with the subtraction of 405 signal from 465-490 signal to extract the calcium-dependent fluorescence signal. An exponential fit algorithm was used to correct for baseline shift. The signal was calculated in dF/F (%) with the equation dF/F = $(F_{fluorescence}-F_{baseline})/F_{baseline}) \times 100$. The Findpeaks algorithm was used to detect SEs with peaks in dF/F greater than 50% of the maximum signal in each recording. Inter-SE intervals represent the time between peak values of each SEs. Multi-peak SE in PNA animals and cluster SE in OVX were analyzed as one SE and SE intervals were calculated from the first peak.

Serial blood sampling, GnRH injection, and LH measurement

Mice were handled daily for at least 3 weeks to habituate for tail-tip bleeding. To examine the relationship between ARN^{KISS} neuron SEs and pulsatile LH secretion, freely behaving mice were attached to the fiber photometry system, and 4 µL blood samples were taken every 6 or 12 min from the tail-tip over a period of 2 to 4 hours during recording. For standard serial blood sampling experiments, LH secretion was measured by obtaining 4 µL blood samples every 10 minutes for 4 hours (between 12:00 and 17:00). At the end of the bleeding period, 200 ng/kg body weight of GnRH (Bachem, catalog No. 4033013, Switzerland) was injected i.p., and two more blood samples were collected at 10-minute intervals. Whole blood samples were diluted 1:15 in 1x PBS with 0.05% Tween 20 (PBS-T) for LH ELISA (Steyn, Wan et al. 2013), an assay sensitivity of 0.04 ng/mL, as well as intra-assay and interassay coefficients of variation of 9.3% and 10.5%, respectively. Pulses were defined and analyzed by PULSAR with the settings optimized for intact female mice: G1 = 3.5, G2 = 2.6, G3 = 1.9, G4 = 1.5, and G5 = 1.2 (Porteous, Haden et al. 2021).

K-means clustering

Data analysis and unsupervised k-means clustering analysis were performed without vaginal cytology information as previously described (Vas, Wall et al. 2024). Synchronization events with multiple peaks occurring within 160s were defined as mpSEs. In brief, 4-day recordings were segmented into 24-hour periods before applying a customized MATLAB code to extract features of the population activity of the ARN^{KISS} neurons including the (i) number of SEs within a given time period; (ii) prevalence of different type of SEs (single- and mpSEs); (iii) inter-SE intervals, variation in the inter-SE intervals (reflecting regularity of activity patterns); and (iv) shape of the SEs (height, variation in height, half-duration). These 24-hour periods were further fragmented using a 7-hour window size with 1-hour sliding steps. The first and the last 3 hours of each 24-hour recording were extrapolated from the assignment of the 4th and 21st hours, respectively. The following parameters were used to describe the activity pattern of ARN^{KISS} neurons: (i) number of SEs, where mpSEs were counted as a single SE; (ii) standard deviation (SD) of inter-SE intervals; (iii) number of mpSEs; (iv) average number of peaks within an mpSEs; and (v) average duration of mpSEs. All these parameter values were then linearly rescaled to values between 0 and 1. To perform k-means clustering on the data set, we used RapidMiner Studio's (*https://rapidminer.com*, educational license) built-in module (k = 5, number of runs with random initiations = 100, measure types = "NumericalMeasures", numerical measure = "EuclideanDistance", max optimization steps = 100). Cluster centroid data and cluster labels (to each 7-hour time window) were exported from RapidMiner Studio and were visualized in R (using "ggplot2", "readr" and "fmsb" libraries).

Progesterone injection

Progesterone stock solution (800 μg/mL) was prepared by dissolving 8 mg of progesterone (Sigma, Catalog No. P0130) in 1 mL ethanol followed by 1:10 dilution in sesame oil. Diestrous female mice were recorded for 3 hours before an i.p. injection of progesterone (4 mg/kg body weight) or vehicle (100 μL, sesame oil). A previous study has shown that i.p. injection of progesterone at 8 mg/kg body weight to C57BL/6 mice results in peak circulating progesterone concentrations of 110 ng/mL before dropping to 34 ng/mL 1 hour later (Wong, Ray et al. 2012). Circulating progesterone levels at proestrus in C57BL/6 mice are 22 ± 4 ng/mL (Wall, Desai et al. 2023), so we employed half of this dose to test the sensitivity of progesterone negative feedback to ARN^{KISS} neurons.

Statistical analysis

All statistical analyses were performed using GraphPad Prism 10 software. Depending on different experimental designs, and the nature of data distribution, data were analyzed by Mann-Whitney U tests, Kruskal-Wallis test followed by Dunn's multiple comparison tests, two-way ANOVA followed by Sidak's multiple comparisons test, or two-way repeated-measures ANOVA followed by Sidak's multiple comparisons test. The threshold level for statistical significance was set at $P<0.05$ with data presented as mean ± SEM with each dot representing one animal.

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Reviewer #2 (Public review):

Summary:

The authors aimed to investigate the functionality of the GnRH (gonadotropin-releasing hormone) pulse generator in different mouse models to understand its role in reproductive physiology and its implications for conditions like polycystic ovary syndrome (PCOS). They compared the GnRH pulse generator activity in control mice, peripubertal androgen (PPA) treated mice, and prenatal androgen (PNA) exposed mice. The study sought to elucidate how androgen exposure affects the GnRH pulse generator and subsequent LH (luteinizing hormone) secretion, contributing to the pathophysiology of PCOS.

Strengths:

(1) Comprehensive Model Selection: The use of both PPA and PNA mouse models allows for a comparative analysis that can distinguish the effects of different timings of androgen exposure.

(2) Detailed Methodology: The methods employed, such as photometry recordings and serial blood sampling, are robust and allow for precise measurement of GnRH pulse generator activity and LH secretion.

(3) Clear Results Presentation: The experimental results are well-documented with appropriate statistical analyses, ensuring the findings are reliable and reproducible.

(4) Relevance to PCOS: The study addresses a significant gap in understanding the neuroendocrine mechanisms underlying PCOS, making the findings relevant to both basic science and potentially clinical research.

Weaknesses

(1) Model Limitations: While the PNA mouse model is suggested as the most appropriate for studying PCOS, the authors acknowledge that it does not completely replicate the human condition, particularly the elevated LH response seen in women with PCOS.

(2) Complex Data Interpretation: The reduced progesterone feedback and its effects on the GnRH pulse generator in PNA mice add complexity to data interpretation, making it challenging to draw straightforward conclusions.

(3) Machine Learning (ML) Selection and Validation: While k-means clustering is a useful tool for pattern recognition, the manuscript lacks detailed justification for choosing this specific algorithm over other potential methods. The robustness of clustering results has not been validated.

(4) Biological Interpretability: Although the machine learning approach identified cyclical patterns, the biological interpretation of these clusters in the context of PCOS is not thoroughly discussed. A deeper exploration of how these clusters correlate with physiological and pathological states could enhance the study's impact.

(5) Sample Size: The study uses a relatively small number of animals (n=4-7 per group), which may limit the generalisability of the findings. Larger sample sizes could provide more robust and statistically significant results.

(6) Scope of Application: The findings, while interesting, are primarily applicable to mouse models. The translation to human physiology requires cautious interpretation and further validation.

Comments on revised version:

I did not find the response to my main concerns regarding justification for the choice of the number of clusters (k) and providing evidence of cluster robustness satisfactory at all. It sounds contradictory to me to state that the authors have used unsupervised ML approach when at the same time had clear understanding of the data and the features they wanted to capture. Unsupervised approaches are meant to reveal features that are not apparent by eye... however in their response the authors state, "...our aim was to develop an unsupervised approach that would automatically detect the onset and existence of the key features of pulse generator cyclicity that were apparent by eye...". This sounds like a rather supervised ML approach to me.
Furthermore, I am still unsure why did the authors choose k=5, i.e. assumed there are 5 clusters in the data, and did they explore other possible values for k?
- If not why not? How does this fit with the claims that their ML approach is unsupervised, in

other words purely data-driven without making any assumptions?
- If yes did they compare the robustness of their clustering results obtained for different values of k?

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Reviewer #3 (Public review):

Summary:

Zhou and colleagues elegantly used pre-clinical mouse models to understand the nature of abnormally high GnRH/LH pulse secretion in polycystic ovary syndrome (PCOS), a major endocrine disorder affecting female fertility worldwide. This work brings a fundamental question of how altered gonadotropin secretion takes place upstream within the GnRH pulse generator core, which is defined by arcuate nucleus kisspeptin neurons.

Strengths:

Authors use state-of-the-art in vivo calcium imaging with fiber photometry and important physiological manipulations and measurements to dissect the possible neuronal mechanisms underlying such neuroendocrine derangements in PCOS. The additional use of unsupervised k-means clustering analysis for the evaluation of calcium synchronous events greatly enhances the quality of their evidence. The authors nicely propose that neuroendocrine dysfunction in PCOS might involve different setpoints through the hypothalamic-pituitary-gonadal (HPG) axis, and beyond kisspeptin neurons, which importantly pushes our field forward toward future investigations.

Weaknesses:

The reviewer agrees that the authors provide important evidence and have improved the quality of the manuscript following first-round revisions. However, they seem resistant to show frequency and amplitude averages in Figure 1 or as supplemental data. Whether the amplitude is dependent on fiber position and its influences on the analysis should be a point of discussion and not data omission. A more detailed analysis of frequency data would enhance the quality of their manuscript.

Comments on revised version:

This comment is related to Reviewer 3's comment # 2 (major) response:

The response does not justify why authors could simply show frequency and amplitude averages in Figure 1 or as supplemental data. Whether the amplitude is dependent on fiber position and its influences on the analysis should be a point of discussion and not data omission.

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Author response:

The following is the authors' response to the previous reviews.

> ***Reviewer #1 (Public Review):***
>
> *The manuscript involves 11 research vignettes that interrogate key aspects of GnRH pulse generator in two established mouse models of PCOS (peripubertal and prenatal androgenisation; PPA and PNA) (9 of the vignettes focus on the latter model).*

> *A key message of this paper is that the oft-quoted idea of rapid GnRH/LH pulses associated with PCOS is in fact not readily demonstrable in PNA and PPA mice. This is an important message to make known, but when established dogmas are being challenged, the experiments behind them need to be robust. In this case, underpowered experiments and one or two other issues greatly limit the overall robustness of the study.*
>
> *General critiques*
>
> *(1) My main concern is that many/most of the experiments were limited to 4-5 mice per group (PPA experiments 1 and 2, PNA experiments 3, 5, 6, 8, and 9). This seems very underpowered for trying to disprove established dogmas (sometimes falling back on "non-significant trends" - lines 105 and 239).*

For the key characterization of GnRH pulse generator activity and LH pulsatility in intact PNA mice (Fig.3, 4, 6), we used 6-8 animals in each experiment which we believe to be sufficient.

It is pertinent to explore the “established dogma”. While there is every expectation that the PNA model should have increased LH pulsatility, in fact there is only a single study (Moore, Prescott et al. 2015) that has shown this. The two other reports that have examined this issue find no change in LH pulse frequency (McCarthy, Dischino et al. 2021 and ours). Hence, we would suggest that expectations rather than evidence presently maintains the PNA “dogma”. For the PPA model, there is in fact not a single paper reporting increased LH pulse frequency.

> *(2) Page 133-142: it is concerning that the PNA mice didn't have elevated testosterone levels, and this clearly isn't the fault of the assay as this was re-tested in the laboratory of Prof Handelsman, an expert in the field, using LCMS. The point (clearly made in lines 315-336 of the Discussion) that elevated testosterone in PNA mice has been shown in some but not other publications is an important concern to describe for the field. However, the fact remains that it IS elevated in numerous studies, and in the current study it is not so, yet the authors go on to present GnRH pulse generator data as characteristic of the PNA model. Perhaps a demonstration of elevated testosterone levels (by LCMS?) should become a standard model validation prerequisite for publishing any PNA model data.*

We provide a Table below showing the huge inconsistencies in testosterone levels reported in the PNA mouse model. If anything, these inconsistencies might be explained by age, although again this is very variable between studies. Much the same as the “dogma” related to LH pulsatility in the PNA model, we would question whether there is any robust increase in testosterone levels in this model. There is no question that women with PCOS have elevated testosterone but whether the PNA mouse is a good model for this is debatable. We have noted this caution and the need for further LC-MS studies in the Discussion.

Author response table 1.

Testosterone levels	Age of mice	Measurement	Source
Increased in PNA	4-6 months	Radioimmunoassay (Diagnostic Products, Los Angeles)	Sullivan and Moenter 2004
	8 months	Radioimmunoassay (Millipore, cat# SRI-13K)	Roland, Nunemaker et al. 2010
	60-80 days (~2-3 months)	ELISA (Demeditec Diagnostics, GmnH)*	Moore, Prescott et al. 2015
	2 & 3 months	ELISA (YANYU, Shanghai, China)	Lei, Ding et al. 2017
	50-60 days (~1.7-2 months)	ELISA (LDN)	Silva, Prescott et al. 2018
No difference	5 months	Radioimmunoassay (Millipore, cat# SRI-13K)	Roland, Nunemaker et al. 2010
	2-4 months	Radioimmunoassay (catalog no. TKTT2; Siemens Medical Solutions, Los Angeles, CA)	Roland and Moenter 2011
	3 weeks (~1 month)	ELISA (YANYU, Shanghai, China)	Lei, Ding et al. 2017
	30-40 days (~1-1.3 months)	ELISA (LDN)	Silva, Prescott et al. 2018

*Same ELISA used in the current study.

> *(3) Line 191-196: the lack of a significant increase in LH pulse frequency in PNA mice is based on measurements using reasonable group sizes (7-8), although the sampling frequency is low for this type of analysis (10-minute intervals; 6-minute intervals would seem safer for not missing some pulses). The significance of the LH pulse frequency results is not stated (looks like about p=0.01). The authors note that LH concentration IS elevated (approximately doubled), and this clearly is not caused by an increase in amplitude (Figure 4 G, H, I). These things are worth commenting on in the discussion.*

We have included the p-value of the LH pulse frequency results and included the relevant discussion.

> *(4) An interesting observation is that PNA mice appear to continue to have cyclical patterns of GnRH pulse generator activity despite reproductive acyclicity as determined by vaginal cytology (lines 209-241). This finding was used to analyse the frequency of GnRH pulse generator SEs in the machine-learning-identified diestrous-like stage of PNA mice and compare it to diestrous control mice (as identified by vaginal cytology?) (lines 245-254). The idea of a cycle stage-specific comparison is good, but surely the only valid comparison would be to use machine-learning to identify the diestrous-like stage in both groups of mice. Why use machine learning for one and vaginal cytology for the other?*

As "machine learning-defined" diestrus is based on the control vaginal cytology information, the diestrous mice are in fact defined by the same machine learning parameters. We have now noted this.

> *Specific points*
>
> *(5) With regard to point 2 above, it would be helpful to note the age at which the testosterone samples were taken.*

We have included the age in the method.

> *(6) Lines 198-205 and 258-266: I think these are repeated measures of ANOVA data? If so, report the main relevant effect before the post hoc test result.*

We have included the relevant main effect in the manuscript.

> *(7) Line 415: I don't think the word "although" works in this sentence.*

We have changed the wording accordingly.

> *(8) Lines 514-518: what are the limits of hormone detection in the LCMS assay?*

These were originally stated in the figure legend but have now been included in the Methods.

> ***Reviewer #2 (Public Review):***
>
> *Summary*
>
> *The authors aimed to investigate the functionality of the GnRH (gonadotropin-releasing hormone) pulse generator in different mouse models to understand its role in reproductive physiology and its implications for conditions like polycystic ovary syndrome (PCOS). They compared the GnRH pulse generator activity in control mice, peripubertal androgen (PPA) treated mice, and prenatal androgen (PNA) exposed mice. The study sought to elucidate how androgen exposure affects the GnRH pulse generator and subsequent LH (luteinizing hormone) secretion, contributing to the pathophysiology of PCOS.*
>
> *Strengths*
>
> *(1) Comprehensive Model Selection: The use of both PPA and PNA mouse models allows for a comparative analysis that can distinguish the effects of different timings of androgen exposure.*
>
> *(2) Detailed Methodology: The methods employed, such as photometry recordings and serial blood sampling, are robust and allow for precise measurement of GnRH pulse generator activity and LH secretion.*
>
> *(3) Clear Results Presentation: The experimental results are well-documented with appropriate statistical analyses, ensuring the findings are reliable and reproducible.*
>
> *(4) Relevance to PCOS: The study addresses a significant gap in understanding the neuroendocrine mechanisms underlying PCOS, making the findings relevant to both basic science and potentially clinical research.*
>
> *Weaknesses*
>
> *(1) Model Limitations: While the PNA mouse model is suggested as the most appropriate for studying PCOS, the authors acknowledge that it does not completely replicate the human condition, particularly the elevated LH response seen in women with PCOS.*

We agree.

> *(2) Complex Data Interpretation: The reduced progesterone feedback and its effects on the GnRH pulse generator in PNA mice add complexity to data interpretation, making it challenging to draw straightforward conclusions.*

We agree.

> *(3) Machine Learning (ML) Selection and Validation: While k-means clustering is a useful tool for pattern recognition, the manuscript lacks detailed justification for choosing this specific algorithm over other potential methods. The robustness of clustering results has not been validated.*

Please see below.

> *(4) Biological Interpretability: Although the machine learning approach identified cyclical patterns, the biological interpretation of these clusters in the context of PCOS is not thoroughly discussed. A deeper exploration of how these clusters correlate with physiological and pathological states could enhance the study's impact.*

It is presently difficult to ascribe specific functions of the various pulse generator states to physiological impact. While it is reasonable to suggest that Cluster_0 activity (representing very infrequent SEs) is responsible for the estrous/luteal-phase pause in pulsatility, we remain unclear on the physiological impact of multi-peak SEs on LH secretion, even in normal mice (see Vas et al., Endo 2024). Thus, for the moment, it is most appropriate to simply state that pulse generator activity remains cyclical in PNA mice without any unfounded speculation.

> *(5) Sample Size: The study uses a relatively small number of animals (n=4-7 per group), which may limit the generalisability of the findings. Larger sample sizes could provide more robust and statistically significant results.*

For the key characterization of GnRH pulse generator activity and LH pulsatility in intact PNA mice (Fig.3, 4, 6), we used 6-8 animals in each experiment which we believe to be sufficient. Some of the subsequent experiments do have smaller N numbers and we are particularly aware of the progesterone treatment study that only has N=3 for the PNA group. However, as this was sufficient to show a statistical difference we did not generate more mice.

(6) Scope of Application: The findings, while interesting, are primarily applicable to mouse models. The translation to human physiology requires cautious interpretation and further validation.

We agree.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *(1) The validation of clustering results through additional metrics or comparison with other algorithms would strengthen the methodology. Specifically, the authors selected k=5 for k-means clustering without providing an explicit rationale or evidence of exploratory data analysis (EDA) to support this choice. They refer to their previous publication (Vas, Wall et al. 2024), which does not provide any EDA regarding the choice of a number of clusters nor their robustness. The arbitrary selection of "k" without justification can undermine confidence in the clustering results since clustering results heavily depend on "k". The authors also choose to use Euclidean distance as the*

"numerical measure" setting in the RapidMiner Studio's software without justification given the chosen features used for clustering and their properties. The lack of exploratory analysis to determine the optimal number of clusters, "k", to be considered means that the authors might have missed identifying the true structure of the data. Common cluster robustness methods, like the elbow method or silhouette analysis, are crucial for justifying the number of clusters. An inappropriate choice could lead to incorrect conclusions about the synchronisation patterns of ARN kisspeptin neurons and their implications for the study's hypotheses. Including EDA and other validation techniques (e.g., silhouette scores, elbow method) would have strengthened the manuscript by providing empirical support for the chosen algorithm and settings.

It is important to clarify that we did not start this exercise with an unknown or uncharacterised data set and that the objective of the clustering was not to provide any initial pattern to the data. Rather, our aim was to develop an unsupervised approach that would automatically detect the onset and existence of the key features of pulse generator cyclicity that were apparent by eye e.g. the estrous stage slowing and the presence of multi-peak SEs in metestrous. As such, our optimization was driven by the data as well as observation while retaining the unsupervised nature of k-means clustering. We started by assessed 10 variables describing all possible features of the recordings and through a process of elimination found that just 5 were sufficient to describe the key stages of the cycle. While we appreciate that the use of multiple different algorithms would progressively increase the robustness of the machine learning approach, it is evident that the current k-means approach with k=5 is already very effective at reporting the estrous cyclicity of the pulse generator in normal mice (Vas et al., Endo 2024). Having validated this approach, we have now used it here to compare the cyclical patterns of activity of PNA- and vehicle-treated mice.

(2) The data and methods presented in this study could be valuable for the research community studying reproductive endocrinology and neuroendocrine disorders provided the authors address my comments above regarding the application of ML methods. The insights gained from this work could potentially inform clinical research aiming to develop better diagnostic and therapeutic strategies for PCOS.

Reviewer #3 (Public Review):

Summary:

Zhou and colleagues elegantly used pre-clinical mouse models to understand the nature of abnormally high GnRH/LH pulse secretion in polycystic ovary syndrome (PCOS), a major endocrine disorder affecting female fertility worldwide. This work brings a fundamental question of how altered gonadotropin secretion takes place upstream within the GnRH pulse generator core, which is defined by arcuate nucleus kisspeptin neurons.

Strengths:

The authors use state-of-the-art in vivo calcium imaging with fiber photometry and important physiological manipulations and measurements to dissect the possible neuronal mechanisms underlying such neuroendocrine derangements in PCOS. The additional use of unsupervised k-means clustering analysis for the evaluation of calcium synchronous events greatly enhances the quality of their evidence. The authors nicely propose that neuroendocrine dysfunction in PCOS might involve different setpoints through the hypothalamic-pituitary-gonadal (HPG) axis, and beyond kisspeptin neurons, which importantly pushes our field forward toward future investigations.

Weaknesses:

> *Although the authors provide important evidence, additional efforts are required to improve the quality of the manuscript and back up their claims. For instance, animal experiments failed to detect high testosterone levels in PNA female mice, a well-established PCOS mouse model. Considering that androgen excess is a hallmark of PCOS, this highly influences the subsequent evaluation of calcium synchronous events in arcuate kisspeptin neurons and the implications for neuroendocrine derangements.*

Please see our response to Reviewer 1. It will be important to establish a robust PCOS mouse model in the future that has elevated pulse generator activity in the presence of elevated testosterone concentrations.

> *Authors also may need to provide LH data from another mouse model used in their work, the peripubertal androgen (PPA) model. Their claims seem to fall short without the pairing evidence of calcium synchronous events in arcuate kisspeptin neurons and LH pulse secretion.*

We have demonstrated that ARN-KISS neuron SEs are perfectly correlated with pulsatile LH secretion in intact and gonadectomized male and female mice on many occasions. Given that the pulse generator frequency slows by 50% in PPA mice, it is very hard to imagine how this could result in an elevated LH pulse frequency. While we were undertaking these studies the first paper (to our knowledge) looking at pulsatile LH secretion in the PPA model was published; no change was found.

> *Another aspect that requires reviewing, is further exploration of their calcium synchronous events data and the increase of animal numbers in some of their experiments.*

Please see below.

> ***Reviewer #3 (Recommendations For The Authors):***
>
> *The reviewer believes that this work will greatly contribute to the field and, to provide better manuscript quality, there might be only a few minor and major revisions to be included in the future version.*
>
> *Minor:*
>
> *(1) Line 17: I would change the sentence to "One in ten women in their reproductive age suffer from PCOS" to adapt to more accurate prevalence studies.*

We have revised the sentence as recommended.

> *(2) Line 18 and 19: Although the evidence indeed points to a high LH pulse secretion in PCOS, I would change it to "with increased LH secretion" as most studies show mean values and not LH pulse release data.*

While we agree that most human studies show a mean increase in LH, when assessed with sufficient temporal resolution, this results from elevated LH pulse frequency. As such, and to keep the manuscript focussed on the pulse generator, we would like the retain the present wording.

> *(3) Line 47: Please correct "polycystic ovaries" to polycystic-like ovarian morphology to adapt to the current AEPCOS guidelines.*

We have revised the sentence as recommended.

> *(4) Line 231: Authors stated that "These PNA mice exhibited a cyclical pattern of activity similar to that of control mice" (Figure 5C and D). Please, include the statistical tests here for this claim. Although they say there aren't differences, the colored fields do not reflect this and seem quite different. Could the authors re-evaluate these claims or provide better examples in the figure?*

We used Sidak's multiple comparisons tests for this analysis (as stated in Results). The key data for assessing overall cyclical activity in PNA and control mice is Fig 5B which suggest very little difference. We accept that the individual traces of activity (Fig.5D) do not look identical to controls and, indeed, they are representative of the data set. The key point is they remain cyclical in an acyclic mouse. We have made sure that this is clear in the text.

> *(5) Subheadings 6 and & of the result section: It sounds confusing to read the foremost claims of the absence of SE differences and next have a clear SE frequency difference in Figures 6 C and D. The reviewer suggests that authors could reorganize the text and figures to make their rationale flow better for future readers.*

We have considered this point carefully but find that re-organization creates its own problems with having to use the machine learning algorithm before describing it. It will always be problematic to incorporate this type of data-reanalysis in an original paper but think this present sequence is the best that can be achieved.

> *(6) Discussion: If PNA female mice did not have elevated testosterone levels, how can the authors compare their results to the current literature? Could this be the case for lacking a more robust ARNKISS neuronal activity output in their experiments? The reviewer recommends a better discussion concerning these aspects.*

Please refer to our response to Reviewer #1 comment (2).

> *(7) Discussion: the authors claim that diestrous PNA mice exhibited highly variable patterns of ARNKISS neuron activity. Would these differences be due to different circulating sex steroid levels or intrinsic properties? Would the inclusion of future in vitro calcium imaging (brain slices) studies contribute to their research question and conclusions? The reviewer recommends a better discussion concerning these aspects.*

We have tried to clarify that the highly variable patterns of activity in "diestrous" PNA mice come from the fact that we are actually randomly recording from ARN-KISS neurons at metestrus, diestrus, proestrus and estrus. The pulse generator is cycling but we only have the acyclic "diestrous" smear to go by. This also makes brain slice studies difficult as we would never know the actual cycle stage.

> *Major:*
>
> *(1) Results section: The reviewer strongly recommends that the LH pulse secretion data for the PPA group be included in the manuscript. If the SEs represent the central mechanism of pulse generation, would the LH pulse frequency match those events? If not, could a mismatch be explained by androgen-mediated negative feedback at the pituitary level? What is the pituitary LH response to exogenous GnRH (i.p. injection) in the PPA group?*

Our initial observation showed the frequency of ARNKISS neuron SEs was halved in PPA mice compared to controls. Additionally, one study reported pulsatile LH secretion to be unchanged in this animal model (Coyle, Prescott et al. 2022). Both pieces of evidence clearly indicate that the PPA mouse does not provide an appropriate PCOS model of elevated pulse generator activity. Therefore, we do not see the value of pursuing further experiments in this animal model.

> *(2) Although the evaluation of relative frequency and normalized amplitude indicate the dynamic over time, the authors should include the average amplitudes and frequencies of events within the recording session. For instance, looking at Figures 1 A and B and Figures 3 A and B, a reader can observe differences in the amplitude due to different scaling axes. Perhaps, using a Python toolbox such as GuPPy or any preferred analysis pipeline might help authors include these parameters.*

The amplitude of recorded SEs for each mouse depends primarily on the fiber position. As such, it has only ever been possible to assess SE amplitude changes within the same mouse. It is not possible to assess differences in SE amplitude between mice.

> *(3) Line 144-156: (Immunoreactivity results): Authors should proceed with caution when describing these results and clearly state that results show a software-based measurement of immunoreactive signal intensity. In addition, the small sample size of the PNA group (N = 4) compared to controls (N = 6-7) seems to mask possible differences. Could the authors increase the N of the PNA group and re-evaluate these results?*

We have clarified that the immunoreactive signal intensity is based on software-based measurement. The N number for PNA mice in these studies varies from 4 to 6 depending on brain section availability for the different immunohistochemistry runs. The scatter of data is such that any new data points would need to be at the extreme of the distributions to likely have any impact on statistical significance. As a minor part of the paper, we did not feel that the use of further mice was warranted.

> *(4) Considering the great variability of PNA's number of SE/hr, the review suggests increasing the N in this group, thus, authors can re-evaluate their findings and draw better analysis/ conclusion.*

We have n=6 for the PNA group in the study. As noted above, the variability in SE/hr in Figure 3 comes from assessing the pulse generator at random times within the estrous cycle. Once we separate “diestrous-like” stage for the PNA animals, the variability is decreased as shown in Figure 6.

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